QLoRA: Efficient Finetuning of Quantized LLMs

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Abstract

We present QLoRA, an efficient finetuning approach that reduces memory usage enough to finetune a 65B parameter model on a single 48GB GPU while preserving full 16-bit finetuning task performance. QLoRA backpropagates gradients through a frozen, 4-bit quantized pretrained language model into Low Rank Adapters (LoRA). Our best model family, which we name **Guanaco**, outperforms all previous openly released models on the Vicuna benchmark, reaching 99.3% of the performance level of ChatGPT while only requiring 24 hours of finetuning on a single GPU. QLoRA introduces a number of innovations to save memory without sacrificing performance: (a) 4-bit NormalFloat (NF4), a new data type that is information theoretically optimal for normally distributed weights (b) Double Quantization to reduce the average memory footprint by quantizing the quantization constants, and (c) Paged Optimizers to manage memory spikes. We use QLoRA to finetune more than 1,000 models, providing a detailed analysis of instruction following and chatbot performance across 8 instruction datasets, multiple model types (LLaMA, T5), and model scales that would be infeasible to run with regular finetuning (e.g. 33B and 65B parameter models). Our results show that QLoRA finetuning on a small high-quality dataset leads to state-of-the-art results, even when using smaller models than the previous SoTA. We provide a detailed analysis of chatbot performance based on both human and GPT-4 evaluations showing that GPT-4 evaluations are a cheap and reasonable alternative to human evaluation. Furthermore, we find that current chatbot benchmarks are not trustworthy to accurately evaluate the performance levels of chatbots. A lemon-picked analysis demonstrates where **Guanaco** fails compared to ChatGPT. We release all of our models and code, including CUDA kernels for 4-bit training.[2]

1 Introduction

Finetuning large language models (LLMs) is a highly effective way to improve their performance, [40, 62, 43, 61, 59, 37] and to add desirable or remove undesirable behaviors [43, 2, 4]. However, finetuning very large models is prohibitively expensive; regular 16-bit finetuning of a LLaMA 65B parameter model [57] requires more than 780 GB of GPU memory. While recent quantization methods can reduce the memory footprint of LLMs [14, 13, 18, 66], such techniques only work for inference and break down during training [65].

We demonstrate for the first time that it is possible to finetune a quantized 4-bit model without any performance degradation. Our method, QLoRA, uses a novel high-precision technique to quantize a pretrained model to 4-bit, then adds a small set of learnable Low-rank Adapter weights [28]

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[2] https://github.com/artidoro/qlora and https://github.com/TimDettmers/bitsandbytes

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QLORA：量化大型语言模型的高效微调

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摘要

我们提出了 QLORA，一种高效的微调方法，它能够显著降低内存使用，使在单个 48GB GPU 上微调 65B 参数模型成为可能，同时保留完整的 16 位微调任务性能。QLORA 将梯度通过冻结的、4 位量化的预训练语言模型反向传播到低秩适配器（LoRA）。我们命名为 Guanaco 的最佳模型系列，在 Vicuna 基准测试中超越了所有之前公开发布的模型，达到了 ChatGPT 性能水平的 99.3%，而且仅需在单个 GPU 上进行 24 小时的微调。QLORA 引入了多项节省内存而不牺牲性能的创新：(a) 4 位 NormalFloat（NF4），一种对于正态分布权重在信息论上最优的新数据类型；(b) 双重量化，通过对量化常数进行量化来减少平均内存占用；(c) 分页优化器，用于管理内存峰值。我们使用 QLORA 微调了超过 1,000 个模型，并提供了详细的分析。

1 引言

微调大型语言模型（LLM）是提高其性能的非常有效的方法，[40, 62, 43, 61, 59, 37] 也可以用来增加期望的行为或去除不希望的行为 [43, 2, 4]。然而，对非常大的模型进行微调成本高昂；以常规16位精度微调一个65B参数的LLaMA模型 [57] 需要超过780 GB的GPU内存。虽然近期的量化方法可以减少LLM的内存占用 [14, 13, 18, 66]，但这些技术仅适用于推理，并且在训练过程中会失效 [65]。

我们首次展示了在没有任何性能下降的情况下微调量化的4位模型是可能的。我们的方法QLORA使用一种新型高精度技术将预训练模型量化为4位，然后添加一小组可学习的低秩适配器权重[28]

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that are tuned by backpropagating gradients through the quantized weights.

QLoRA reduces the average memory requirements of finetuning a 65B parameter model from >780GB of GPU memory to <48GB without degrading the runtime or predictive performance compared to a 16-bit fully finetuned baseline. This marks a significant shift in accessibility of LLM finetuning: now the largest publicly available models to date finetunable on a single GPU. Using QLoRA, we train the **Guanaco** family of models, with the second best model reaching 97.8% of the performance level of ChatGPT on the Vicuna [10] benchmark, while being trainable in less than 12 hours on a single consumer GPU; using a single professional GPU over 24 hours we achieve 99.3% with our largest model, essentially closing the gap to ChatGPT on the Vicuna benchmark. When deployed, our smallest **Guanaco** model (7B parameters) requires just 5 GB of memory and outperforms a 26 GB Alpaca model by more than 20 percentage points on the Vicuna benchmark (Table 6).

Table 1: Elo ratings for a competition between models, averaged for 10,000 random initial orderings. The winner of a match is determined by GPT-4 which declares which response is better for a given prompt of the the Vicuna benchmark. 95% confidence intervals are shown ($\pm$). After GPT-4, Guanaco 33B and 65B win the most matches, while Guanaco 13B scores better than Bard.

Model	Size	Elo
GPT-4	-	1348 ± 1
Guanaco 65B	41 GB	1022 ± 1
Guanaco 33B	21 GB	992 ± 1
Vicuna 13B	26 GB	974 ± 1
ChatGPT	-	966 ± 1
Guanaco 13B	10 GB	916 ± 1
Bard	-	902 ± 1
Guanaco 7B	6 GB	879 ± 1

QLoRA introduces multiple innovations designed to reduce memory use without sacrificing performance: (1) **4-bit NormalFloat**, an information theoretically optimal quantization data type for normally distributed data that yields better empirical results than 4-bit Integers and 4-bit Floats. (2) **Double Quantization**, a method that quantizes the quantization constants, saving an average of about 0.37 bits per parameter (approximately 3 GB for a 65B model). (3) **Paged Optimizers**, using NVIDIA unified memory to avoid the gradient checkpointing memory spikes that occur when processing a mini-batch with a long sequence length. We combine these contributions into a better tuned LoRA approach that includes adapters at every network layer and thereby avoids almost all of the accuracy tradeoffs seen in prior work.

QLoRA's efficiency enables us to perform an in-depth study of instruction finetuning and chatbot performance on model scales that would be impossible using regular finetuning due to memory overhead. Therefore, we train more than 1,000 models across several instruction tuning datasets, model architectures, and sizes between 80M to 65B parameters. In addition to showing that QLoRA recovers 16-bit performance (§4) and training a state-of-the-art chatbot, **Guanaco**, (§5), we also analyze trends in the trained models. First, we find that data quality is far more important than dataset size, e.g., a 9k sample dataset (OASST1) outperformed a 450k sample dataset (FLAN v2, subsampled) on chatbot performance, even when both are meant to support instruction following generalization. Second, we show that strong Massive Multitask Language Understanding (MMLU) benchmark performance does not imply strong Vicuna chatbot benchmark performance and vice versa—in other words, dataset suitability matters more than size for a given task.

Furthermore, we also provide a extensive analysis of chatbot performance that uses both human raters and GPT-4 for evaluation. We use tournament-style benchmarking where models compete against each other in matches to produce the best response for a given prompt. The winner of a match is judged by either GPT-4 or human annotators. The tournament results are aggregated into Elo scores [16, 17] which determine the ranking of chatbot performance. We find that GPT-4 and human evaluations largely agree on the rank of model performance in the tournaments, but we also find there are instances of strong disagreement. As such, we highlight that model-based evaluation while providing a cheap alternative to human-annotation also has its uncertainties.

We augment our chatbot benchmark results with a qualitative analysis of **Guanaco** models. Our analysis highlights success and failure cases that were not captured by the quantitative benchmarks.

We release all model generations with human and GPT-4 annotations to facilitate further study. We open-source our codebase and CUDA kernels and integrate our methods into the Hugging Face transformers stack [64], making them easily accessible to all. We release a collection of adapters for 7/13/33/65B size models, trained on 8 different instruction following datasets, for a total of 32 different open sourced, finetuned models.

通过将梯度反向传播到量化权重来调谐的。

QLORA 将微调一个 65B 参数模型的平均显存需求从 >780GB 降低到 <48GB，而不会降低运行时性能或预测性能，相比于 16 位完全微调的基线。这标志着 LLM 微调可及性的重大转变：现在迄今为止最大的公开可用模型可以在单 GPU 上进行微调。使用 QLORA，我们训练了 Guanaco 系列模型，其中第二好的模型在 Vicuna [10] 基准上达到了 ChatGPT 性能水平的 97.8%，而在单个消费级 GPU 上训练时间少于 12 小时；使用单个专业 GPU 进行 24 小时训练，我们最大的模型达到 99.3%，实际上在 Vicuna 基准上缩小了与 ChatGPT 的差距。部署时，我们最小的 Guanaco 模型（7B 参数）只需要 5 GB 内存，并且在 Vicuna 基准测试中比 26 GB 的 Alpaca 模型高出 20 个百分点以上（表 6）。

表1：模型之间比赛的Elo评分，基于10,000次随机初始排序的平均值。比赛的胜者由GPT-4确定，GPT-4会为给定的Vicuna基准测试的提示判断哪一个回应更佳。显示95%的置信区间（±）。在GPT-4之后，Guanaco 33B和65B赢得了最多比赛，而Guanaco 13B的得分高于Bard。

Model	Size	Elo
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Bard	-	902 ± 1
Guanaco 7B	6 GB	879 ± 1

QLORA 引入了多项创新设计，旨在在不牺牲性能的前提下减少内存使用：(1) 4-bit NormalFloat，一种针对正态分布数据的信息论上最优量化数据类型，实证结果优于 4-bit 整数和 4-bit 浮点数。(2) 双重量化，一种对量化常数进行量化的方法，每个参数平均节省约 0.37 位（对于 65B 模型约为 3 GB）。(3) 分页优化器，使用 NVIDIA 统一内存来避免在处理长序列长度的小批量时出现的梯度检查点内存峰值。我们将这些贡献结合到一个经过更好调优的 LoRA 方法中，在每个网络层都包含适配器，从而几乎避免了以往工作中出现的性能精度折衷。

QLORA 的高效性使我们能够在常规微调因内存开销而无法实现的模型规模上，对指令微调和聊天机器人性能进行深入研究。因此，我们在多个指令微调数据集、模型架构以及参数规模从 8000 万到 650 亿的模型上训练了超过 1000 个模型。除了展示 QLORA 如何恢复 16 位性能 (§4) 并训练出最先进的聊天机器人 Guanaco (§5) 外，我们还分析了训练模型中的趋势。首先，我们发现数据质量远比数据集规模更为重要，例如，一个 9 千样本的数据集 (OASST1) 在聊天机器人性能上优于一个 45 万样本的数据集 (FLAN v2，子采样)，即使两者都旨在支持指令遵循的泛化。其次，我们表明，强大的大规模多任务语言理解 (MMLU) 基准性能并不意味着在 Vicuna 聊天机器人基准上的强性能，反之亦然——换句话说，数据集的适用性更为重要。

此外，我们还提供了对聊天机器人性能的广泛分析，该分析使用了人工评分员和 GPT-4 进行评估。我们使用锦标赛式基准测试，模型在比赛中相互竞争，以为给定的提示生成最佳回应。比赛的胜者由 GPT-4 或人工注释员评判。锦标赛结果被汇总为 Elo 分数 [16, 17]，以确定聊天机器人的性能排名。我们发现，在锦标赛中，GPT-4 和人工评估在模型性能排名上大体一致，但我们也发现存在严重分歧的情况。因此，我们强调，基于模型的评估虽然提供了比人工注释更便宜的替代方案，但也存在不确定性。

我们通过对Guanaco模型的定性分析来补充我们的聊天机器人基准测试结果。我们的分析突出了定量基准测试未能捕捉的成功与失败案例。

我们发布了所有模型生成内容，并附有人类和 GPT-4 的标注，以便进行进一步研究。我们开源了我们的代码库和 CUDA 内核，并将我们的方法集成到 Hugging Face transformers 栈中 [64]，使其对所有人都易于访问。我们发布了一系列适配器，适用于 7/13/33/65B 规模的模型，这些适配器在 8 个不同的指令跟随数据集上进行训练，总共提供了 32 个不同的开源微调模型。

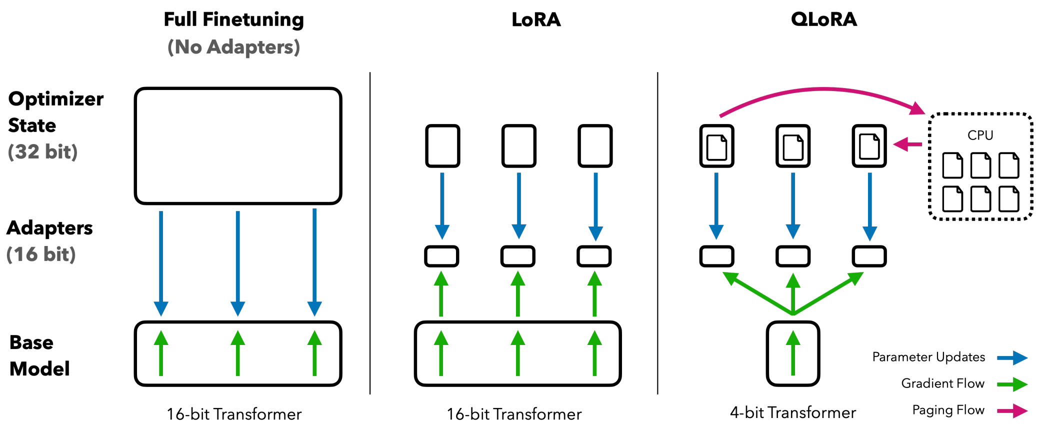

Figure 1: Different finetuning methods and their memory requirements. QLORA improves over LoRA by quantizing the transformer model to 4-bit precision and using paged optimizers to handle memory spikes.

2 Background

Block-wise k-bit Quantization Quantization is the process of discretizing an input from a representation that holds more information to a representation with less information. It often means taking a data type with more bits and converting it to fewer bits, for example from 32-bit floats to 8-bit Integers. To ensure that the entire range of the low-bit data type is used, the input data type is commonly rescaled into the target data type range through normalization by the absolute maximum of the input elements, which are usually structured as a tensor. For example, quantizing a 32-bit Floating Point (FP32) tensor into a Int8 tensor with range $[-127, 127]$:

$$\mathbf{X}^{\text{Int8}} = \text{round}\left(\frac{127}{\text{absmax}(\mathbf{X}^{\text{FP32}})}\mathbf{X}^{\text{FP32}}\right) = \text{round}(c^{\text{FP32}} \cdot \mathbf{X}^{\text{FP32}}), \tag{1}$$

where c is the *quantization constant* or *quantization scale*. Dequantization is the inverse:

$$\text{dequant}(c^{\text{FP32}}, \mathbf{X}^{\text{Int8}}) = \frac{\mathbf{X}^{\text{Int8}}}{c^{\text{FP32}}} = \mathbf{X}^{\text{FP32}} \tag{2}$$

The problem with this approach is that if a large magnitude value (i.e., an outlier) occurs in the input tensor, then the quantization bins—certain bit combinations—are not utilized well with few or no numbers quantized in some bins. To prevent the outlier issue, a common approach is to chunk the input tensor into blocks that are independently quantized, each with their own quantization constant c. This can be formalized as follows: We chunk the input tensor $\mathbf{X} \in \mathbb{R}^{b \times h}$ into n contiguous blocks of size B by flattening the input tensor and slicing the linear segment into $n = (b \times h)/B$ blocks. We quantize these blocks independently with Equation 1 to create a quantized tensor and n quantization constants c_i.

Low-rank Adapters Low-rank Adapter (LoRA) finetuning [28] is a method that reduces memory requirements by using a small set of trainable parameters, often termed adapters, while not updating the full model parameters which remain fixed. Gradients during stochastic gradient descent are passed through the fixed pretrained model weights to the adapter, which is updated to optimize the loss function. LoRA augments a linear projection through an additional factorized projection. Given a projection $\mathbf{XW} = \mathbf{Y}$ with $\mathbf{X} \in \mathbb{R}^{b \times h}$, $\mathbf{W} \in \mathbb{R}^{h \times o}$ LoRA computes:

$$\mathbf{Y} = \mathbf{XW} + s\mathbf{XL}_1\mathbf{L}_2, \tag{3}$$

where $\mathbf{L}_1 \in \mathbb{R}^{h \times r}$ and $\mathbf{L}_2 \in \mathbb{R}^{r \times o}$, and s is a scalar.

Memory Requirement of Parameter-Efficient Finetuning One important point of discussion is the memory requirement of LoRA during training both in terms of the number and size of adapters used. Since the memory footprint of LoRA is so minimal, we can use more adapters to improve performance without significantly increasing the total memory used. While LoRA was designed as a

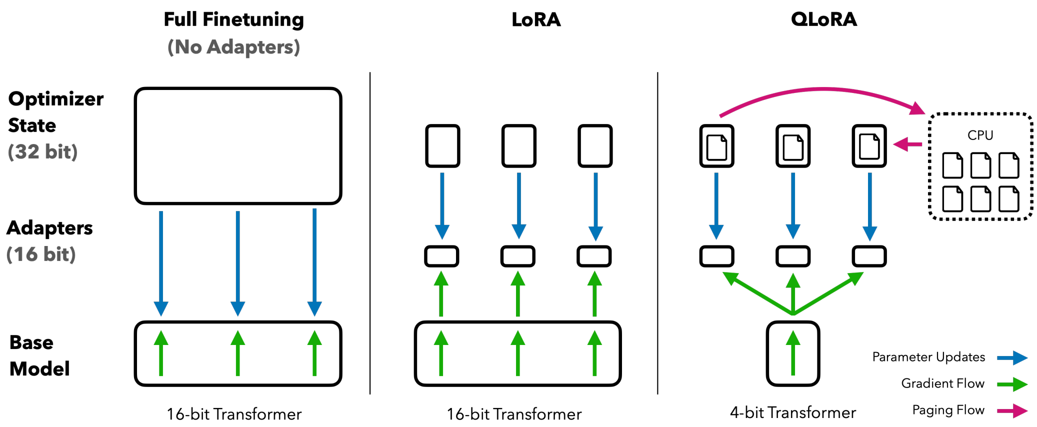

图 1：不同微调方法及其内存需求。QLORA 通过将变换器模型量化为 4 位精度并使用分页优化器来处理内存峰值，从而优于 LoRA。

2 背景

按块 k 位量化 量化是将输入从信息量更大的表示离散化为信息量较少的表示的过程。它通常意味着将更多位的数据类型转换为更少位的数据类型，例如从 32 位浮点数转换为 8 位整数。为了确保使用低位数据类型的整个范围，通常通过对输入元素的绝对最大值进行归一化，将输入数据类型重新缩放到目标数据类型范围，这些输入元素通常以张量的形式组织。例如，将 32 位浮点（FP32）张量量化为范围为 $[-127, 127]$ 的 Int8 张量:

$$\mathbf{X}^{\text{Int8}} = \text{round}\left(\frac{127}{\text{absmax}(\mathbf{X}^{\text{FP32}})}\mathbf{X}^{\text{FP32}}\right) = \text{round}(c^{\text{FP32}} \cdot \mathbf{X}^{\text{FP32}}), \tag{1}$$

其中 c 是 *qua ntization constant* 或 *quantization scale*。去量化是 逆

$$\text{dequant}(c^{\text{FP32}}, \mathbf{X}^{\text{Int8}}) = \frac{\mathbf{X}^{\text{Int8}}}{c^{\text{FP32}}} = \mathbf{X}^{\text{FP32}} \tag{2}$$

这种方法的问题在于，如果输入张量中出现一个大幅值（即异常值），那么量化区间——某些比特组合——就不会被充分利用，某些区间可能量化的数值很少或没有数值。为防止异常值问题，一种常见的方法是将输入张量分块，每块独立量化，每块都有自己的量化常数 c。可以形式化如下：我们将输入张量 $\mathbf{X} \in \mathbb{R}^{b\times h}$ 展平并切分成 $n = (b \times h)/B$ 个连续块，每块大小为 B，形成 n 个块。我们使用公式 1 独立量化这些块，创建一个量化后的张量和 n 个量化常数 c_i。

低秩适配器 低秩适配器（LoRA）微调 [28] 是一种通过使用一小部分可训练参数（通常称为适配器）来减少内存需求的方法，同时不更新保持固定的完整模型参数。在随机梯度下降过程中，梯度通过固定的预训练模型权重传递到适配器，适配器被更新以优化损失函数。LoRA 通过额外的分解投影增强线性投影。给定一个带有 $\mathbf{X} \in \mathbb{R}^{b\times h}$, $\mathbf{W} \in \mathbb{R}^{h\times o}$ 的投影 $\mathbf{XW} = \mathbf{Y}$，LoRA 计算如下:

$$\mathbf{Y} = \mathbf{XW} + s\mathbf{XL}_1\mathbf{L}_2, \tag{3}$$

其中 $\mathbf{L}_1 \in \mathbb{R}^{h\times r}$ 和 $\mathbf{L}_2 \in \mathbb{R}^{r\times o}$，并且 s 是一个标量。

参数高效微调的内存需求 一个重要的讨论点是 LoRA 在训练过程中对内存的需求，包括所使用的适配器的数量和大小。由于 LoRA 的内存占用非常小，我们可以使用更多的适配器来提高性能，而不会显著增加总内存使用量。虽然 LoRA 是被设计为

Parameter Efficient Finetuning (PEFT) method, most of the memory footprint for LLM finetuning comes from activation gradients and not from the learned LoRA parameters. For a 7B LLaMA model trained on FLAN v2 with a batch size of 1, with LoRA weights equivalent to commonly used 0.2% of the original model weights[28, 37], the LoRA input gradients have a memory footprint of 567 MB while the LoRA parameters take up only 26 MB. With gradient checkpointing [9], the input gradients reduce to an average of 18 MB per sequence making them more memory intensive than all LoRA weights combined. In comparison, the 4-bit base model consumes 5,048 MB of memory. This highlights that gradient checkpointing is important but also that aggressively reducing the amount of LoRA parameter yields only minor memory benefits. This means we can use more adapters without significantly increasing the overall training memory footprint (see Appendix G for a detailed breakdown). As discussed later, this is crucial for recovering full 16-bit precision performance.

3 QLoRA Finetuning

QLoRA achieves high-fidelity 4-bit finetuning via two techniques we propose—4-bit NormalFloat (NF4) quantization and Double Quantization. Additionally, we introduce Paged Optimizers, to prevent memory spikes during gradient checkpointing from causing out-of-memory errors that have traditionally made finetuning on a single machine difficult for large models.

QLoRA has one low-precision storage data type, in our case usually 4-bit, and one computation data type that is usually BFloat16. In practice, this means whenever a QLoRA weight tensor is used, we dequantize the tensor to BFloat16, and then perform a matrix multiplication in 16-bit.

We now discuss the components of QLoRA followed by a formal definition of QLoRA.

4-bit NormalFloat Quantization The NormalFloat (NF) data type builds on Quantile Quantization [15] which is an information-theoretically optimal data type that ensures each quantization bin has an equal number of values assigned from the input tensor. Quantile quantization works by estimating the quantile of the input tensor through the empirical cumulative distribution function.

The main limitation of quantile quantization is that the process of quantile estimation is expensive. Therefore fast quantile approximation algorithms, such as SRAM quantiles [15], are used to estimate them. Due to the approximate nature of these quantile estimation algorithms, the data type has large quantization errors for outliers, which are often the most important values.

Expensive quantile estimates and approximation errors can be avoided when input tensors come from a distribution fixed up to a quantization constant. In such cases, input tensors have the same quantiles making exact quantile estimation computationally feasible.

Since pretrained neural network weights usually have a zero-centered normal distribution with standard deviation σ (see Appendix F), we can transform all weights to a single fixed distribution by scaling σ such that the distribution fits exactly into the range of our data type. For our data type, we set the arbitrary range $[-1, 1]$. As such, both the quantiles for the data type and the neural network weights need to be normalized into this range.

The information theoretically optimal data type for zero-mean normal distributions with arbitrary standard deviations σ in the range $[-1, 1]$ is computed as follows: (1) estimate the $2^k + 1$ quantiles of a theoretical $N(0, 1)$ distribution to obtain a k-bit quantile quantization data type for normal distributions, (2) take this data type and normalize its values into the $[-1, 1]$ range, (3) quantize an input weight tensor by normalizing it into the $[-1, 1]$ range through absolute maximum rescaling.

Once the weight range and data type range match, we can quantize as usual. Step (3) is equivalent to rescaling the standard deviation of the weight tensor to match the standard deviation of the k-bit data type. More formally, we estimate the 2^k values q_i of the data type as follows:

$$q_i = \frac{1}{2}\left(Q_X\left(\frac{i}{2^k+1}\right) + Q_X\left(\frac{i+1}{2^k+1}\right)\right), \tag{4}$$

where $Q_X(\cdot)$ is the quantile function of the standard normal distribution $N(0, 1)$. A problem for a symmetric k-bit quantization is that this approach does not have an exact representation of zero, which is an important property to quantize padding and other zero-valued elements with no error. To

参数高效微调（PEFT）方法中，大型语言模型（LLM）微调的大部分内存占用来自激活梯度，而不是学习到的 LoRA 参数。对于在 FLAN v2 上训练的 7B LLaMA 模型，批量大小为 1，LoRA 权重相当于常用的原模型权重的 0.2%[28, 37]时，LoRA 输入梯度的内存占用为 567 MB，而 LoRA 参数仅占用 26 MB。使用梯度检查点 [9] 后，输入梯度平均降至每个序列 18 MB，使其比所有 LoRA 权重的总和更占用内存。相比之下，4-bit 基模型则消耗 5,048 MB 内存。这强调了梯度检查点的重要性，但同时也说明积极减少 LoRA 参数的数量仅能带来微小的内存收益。这意味着我们可以使用更多的适配器，而不会显著增加整体训练内存占用（详见附录 G 的详细分解）。如后文讨论，这对于...至关重要。

3 QLORA 微调

QLORA 通过我们提出的两种技术——4 位 NormalFloat (NF4) 量化和双重量化，实现了高保真 4 位微调。此外，我们引入了分页优化器，以防止在梯度检查点过程中内存激增，从而导致传统上在单机上对大型模型进行微调时容易出现的内存不足错误。

QLORA 有一种低精度存储数据类型，在我们的情况下通常是 4 位，以及一种计算数据类型，通常是 BFloat16。实际上，这意味着每当使用 QLORA 权重张量时，我们会将张量反量化为 BFloat16，然后在 16 位下执行矩阵乘法。

我们现在讨论 QLORA的组成部分，随后是正式定义 关于 QLORA。

4位NormalFloat量化 NormalFloat（NF）数据类型建立在分位数量化[15]的基础上，分位数量化是一种信息理论上最优的数据类型，可确保每个量化区间从输入张量中分配到相等数量的值。分位数量化的工作原理是通过经验累积分布函数估计输入张量的分位数。

分位数量化的主要限制是分位数估计过程成本很高。因此，使用快速分位数近似算法，如SRAM分位数 [15]，来进行估计。由于这些分位数估计算法的近似性质，该数据类型对异常值的量化误差较大，而异常值通常是最重要的值。

当输入张量来自一个固定到量化常数的分布时，可以避免昂贵的分位数估计和近似误差。在这种情况下，输入张量具有相同的分位数，使得精确的分位数估计在计算上是可行的。

由于预训练神经网络的权重通常具有以零为中心、标准差为 σ (（参见附录 F)）的正态分布，我们可以通过缩放 σ 将所有权重转换为单一的固定分布，使该分布正好适合我们数据类型的范围。对于我们的数据类型，我们将任意范围设置为 $[-1, 1]$。因此，数据类型和神经网络权重的分位数都需要归一化到该范围内。

对于零均值且标准差任意的正态分布 σ，位于 $[-1, 1]$ 范围内的信息理论最优数据类型计算方法如下： (1) 估计理论 $N(0, 1)$ 分布的 $2^k + 1$ 分位数，以获得适用于正态分布的 k 位分位量化数据类型，(2) 将该数据类型的值归一化到 $[-1, 1]$ 范围，(3) 通过绝对最大值重缩放，将输入权重张量归一化到 $[-1, 1]$ 范围进行量化。

一旦权重范围和数据类型范围匹配，我们就可以像往常一样进行量化。步骤（3）等同于将权重张量的标准差重新缩放，以匹配 k 位数据类型的标准差。更正式地，我们如下估计该数据类型的 2^k 个值 q_i：

$$q_i = \frac{1}{2}\left(Q_X\left(\frac{i}{2^k+1}\right) + Q_X\left(\frac{i+1}{2^k+1}\right)\right), \tag{4}$$

其中 $Q_X(\cdot)$ 是标准正态分布 $N(0, 1)$ 的分位数函数。对对称 k 位量化来说，一个问题是这种方法没有对零的精确表示，而零的精确表示是量化填充和其他零值元素而不出错的重要属性。

ensure a discrete zeropoint of 0 and to use all 2^k bits for a k-bit datatype, we create an asymmetric data type by estimating the quantiles q_i of two ranges q_i: 2^{k-1} for the negative part and $2^{k-1}+1$ for the positive part and then we unify these sets of q_i and remove one of the two zeros that occurs in both sets. We term the resulting data type that has equal expected number of values in each quantization bin *k-bit NormalFloat* (NFk), since the data type is information-theoretically optimal for zero-centered normally distributed data. The exact values of this data type can be found in Appendix E.

Double Quantization We introduce *Double Quantization* (DQ), the process of quantizing the quantization constants for additional memory savings. While a small blocksize is required for precise 4-bit quantization [13], it also has a considerable memory overhead. For example, using 32-bit constants and a blocksize of 64 for $\mathbf{W}$, quantization constants add $32/64 = 0.5$ bits per parameter on average. Double Quantization helps reduce the memory footprint of quantization constants.

More specifically, Double Quantization treats quantization constants c_2^{FP32} of the first quantization as inputs to a second quantization. This second step yields the quantized quantization constants c_2^{FP8} and the second level of quantization constants c_1^{FP32}. We use 8-bit Floats with a blocksize of 256 for the second quantization as no performance degradation is observed for 8-bit quantization, in line with results from Dettmers and Zettlemoyer [13]. Since the c_2^{FP32} are positive, we subtract the mean from c_2 before quantization to center the values around zero and make use of symmetric quantization. On average, for a blocksize of 64, this quantization reduces the memory footprint per parameter from $32/64 = 0.5$ bits, to $8/64 + 32/(64 \cdot 256) = 0.127$ bits, a reduction of 0.373 bits per parameter.

Paged Optimizers use the NVIDIA unified memory [3] feature wich does automatic page-to-page transfers between the CPU and GPU for error-free GPU processing in the scenario where the GPU occasionally runs out-of-memory. The feature works like regular memory paging between CPU RAM and the disk. We use this feature to allocate paged memory for the optimizer states which are then automatically evicted to CPU RAM when the GPU runs out-of-memory and paged back into GPU memory when the memory is needed in the optimizer update step.

QLoRA. Using the components described above, we define QLoRA for a single linear layer in the quantized base model with a single LoRA adapter as follows:

$$\mathbf{Y}^{\text{BF16}} = \mathbf{X}^{\text{BF16}}\text{doubleDequant}(c_1^{\text{FP32}}, c_2^{\text{k-bit}}, \mathbf{W}^{\text{NF4}}) + \mathbf{X}^{\text{BF16}}\mathbf{L}_1^{\text{BF16}}\mathbf{L}_2^{\text{BF16}}, \tag{5}$$

where doubleDequant$(\cdot)$ is defined as:

$$\text{doubleDequant}(c_1^{\text{FP32}}, c_2^{\text{k-bit}}, \mathbf{W}^{\text{k-bit}}) = \text{dequant}(\text{dequant}(c_1^{\text{FP32}}, c_2^{\text{k-bit}}), \mathbf{W}^{\text{4bit}}) = \mathbf{W}^{\text{BF16}}, \tag{6}$$

We use NF4 for $\mathbf{W}$ and FP8 for c_2. We use a blocksize of 64 for $\mathbf{W}$ for higher quantization precision and a blocksize of 256 for c_2 to conserve memory.

For parameter updates only the gradient with respect to the error for the adapters weights $\frac{\partial E}{\partial \mathbf{L}_i}$ are needed, and not for 4-bit weights $\frac{\partial E}{\partial \mathbf{W}}$. However, the calculation of $\frac{\partial E}{\partial \mathbf{L}_i}$ entails the calculation of $\frac{\partial \mathbf{X}}{\partial \mathbf{W}}$ which proceeds via equation (5) with dequantization from storage $\mathbf{W}^{\text{NF4}}$ to computation data type $\mathbf{W}^{\text{BF16}}$ to calculate the derivative $\frac{\partial \mathbf{X}}{\partial \mathbf{W}}$ in BFloat16 precision.

To summarize, QLoRA has one storage data type (usually 4-bit NormalFloat) and a computation data type (16-bit BrainFloat). We dequantize the storage data type to the computation data type to perform the forward and backward pass, but we only compute weight gradients for the LoRA parameters which use 16-bit BrainFloat.

4 QLoRA vs. Standard Finetuning

We have discussed how QLoRA works and how it can significantly reduce the required memory for finetuning models. The main question now is whether QLoRA can perform as well as full-model finetuning. Furthermore, we want to analyze the components of QLoRA including the impact of NormalFloat4 over standard Float4. The following sections will discuss the experiments that aimed at answering these questions.

[3] https://docs.nvidia.com/cuda/cuda-c-programming-guide

为了确保零点为离散的0，并且将所有2^k位用于一个k位数据类型，我们通过估计两个范围q_i的分位数q_i来创建一个非对称数据类型：负值部分为2^{k-1}，正值部分为2^{k-1} + - 1，然后我们统一这些q_i集合，并去掉两个集合中都出现的一个零。我们称这种在每个量化区间中期望值相等的数据类型为*k-bit NormalFloat* (NFk)，因为该数据类型在信息论上对以零为中心的正态分布数据是最优的。此数据类型的精确值可在附录E中找到。

双重量化 我们介绍了 *Double Quantization* (DQ)，即对量化常数进行量化以实现额外的内存节省的过程。虽然精确的 4 位量化 [13] 需要较小的块大小，但它也会带来相当大的内存开销。例如，对于 $\mathbf{W}$，使用 32 位常数和块大小为 64 时，量化常数平均每个参数增加 32/64 = 0.5 位。双重量化有助于减少量化常数的内存占用。

更具体地说，双重量化将第一次量化的量化常数 c_2^{FP32} 作为第二次量化的输入。第二步产生了量化后的量化常数 c_2^{FP8} 和第二级量化常数 c_1^{FP32}。我们使用块大小为 256 的 8 位浮点数进行第二次量化，因为 8 位量化不会导致性能下降，这与 Dettmers 和 Zettlemoyer [13] 的结果一致。由于 c_2^{FP32} 为正，我们在量化前从 c_2 中减去均值，以使值围绕零集中并使用对称量化。平均而言，对于块大小为 64，这种量化将每个参数的内存占用从 $32/64 = 0.5$ 位减小到 $8/64 + 32/(64 \cdot 256) = 0.127$ 位，每个参数减少 0.373 位。

分页优化器使用 NVIDIA 统一内存 [3] 功能，该功能在 GPU 偶尔出现内存不足的情况下，自动在 CPU 和 GPU 之间进行页到页的传输，以实现无错误的 GPU 处理。该功能的工作方式类似于 CPU 内存和磁盘之间的常规内存分页。我们使用此功能为优化器状态分配分页内存，当 GPU 内存不足时，这些内存会自动被驱逐到 CPU 内存中，而在优化器更新步骤需要使用时，又会分页回到 GPU 内存中。

QLORA。使用上述描述的组件，我们将单层量化基础模型中的单个线性层与单个LoRA适配器的QLORA定义如下：

$$\mathbf{Y}^{\text{BF16}} = \mathbf{X}^{\text{BF16}}\text{doubleDequant}(c_1^{\text{FP32}}, c_2^{\text{k-bit}}, \mathbf{W}^{\text{NF4}}) + \mathbf{X}^{\text{BF16}}\mathbf{L}_1^{\text{BF16}}\mathbf{L}_2^{\text{BF16}}, \tag{5}$$

其中 doubleDequant(·) 定义为：

$$\text{doubleDequant}(c_1^{\text{FP32}}, c_2^{\text{k-bit}}, \mathbf{W}^{\text{k-bit}}) = \text{dequant}(\text{dequant}(c_1^{\text{FP32}}, c_2^{\text{k-bit}}), \mathbf{W}^{\text{4bit}}) = \mathbf{W}^{\text{BF16}}, \tag{6}$$

我们对 $\mathbf{W}$ 使用 NF4，对 c_2 使用 FP8。我们对 $\mathbf{W}$ 使用 64 的块大小以获得更高的量化精度，对 c_2 使用 256 的块大小以节省内存。

对于参数更新，仅需计算适配器权重 $\frac{\partial E}{\partial \mathbf{L}_i}$ 的误差梯度，而不需要计算 4 位权重 $\frac{\partial E}{\partial \mathbf{W}}$ 的梯度。然而，$\frac{\partial E}{\partial \mathbf{L}_i}$ 的计算涉及 $\frac{\partial \mathbf{X}}{\partial \mathbf{W}}$ 的计算，该计算通过公式 (5) 进行，并将存储数据 $\mathbf{W}^{\text{NF4}}$ 反量化为计算数据类型 $\mathbf{W}^{\text{BF16}}$，以 BFloat16 精度计算导数 $\frac{\partial \mathbf{X}}{\partial \mathbf{W}}$。

总而言之，QLORA 有一种存储数据类型（通常是 4 位的 NormalFloat）和一种计算数据类型（16 位的 BrainFloat）。我们将存储数据类型反量化为计算数据类型以执行前向和反向传递，但我们只计算使用 16 位 BrainFloat 的 LoRA 参数的权重梯度。

4 QLoRA 与标准微调

我们已经讨论了 QLoRA 的工作原理以及它如何显著减少微调模型所需的内存。现在的主要问题是 QLoRA 是否能够与全模型微调一样表现良好。此外，我们希望分析 QLoRA 的各个组成部分，包括 NormalFloat4 对比标准 Float4 的影响。以下各节将讨论旨在回答这些问题的实验。

[3] https://docs.nvidia.com/cuda/cuda-c-programming-guide

Experimental setup. We consider three architectures (encoder, encoder-decoder, and decoder only) and compare QLoRA with 16-bit adapter-finetuning and with full-finetuning for models up to 3B. Our evaluations include GLUE [58] with RoBERTa-large [38], Super-NaturalInstructions (TKInstruct) [61] with T5 [49], and 5-shot MMLU [24] after finetuning LLaMA on Flan v2 [39] and Alpaca [55]. To additionally study the advantages of NF4 over other 4-bit data types, we use the setup of Dettmers and Zettlemoyer [13] and measure post-quantization zero-shot accuracy and perplexity across different models (OPT [72], LLaMA [57], BLOOM [52], Pythia [7]) for model sizes 125m - 13B. We provide more details in the results section for each particular setup to make the results more readable. Full details in Appendix A.

While paged optimizers are critical to do 33B/65B QLoRA tuning on a single 24/48GB GPU, we do not provide hard measurements for Paged Optimizers since the paging only occurs when processing mini-batches with long sequence lengths, which is rare. We do, however, perform an analysis of the runtime of paged optimizers for 65B models on 48GB GPUs and find that with a batch size of 16, paged optimizers provide the same training speed as regular optimizers. Future work should measure and characterize under what circumstances slowdowns occur from the paging process.

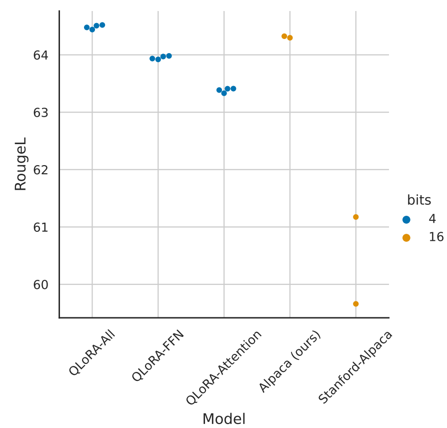

Figure 2: RougeL for LLaMA 7B models on the Alpaca dataset. Each point represents a run with a different random seed. We improve on the Stanford Alpaca fully finetuned default hyperparameters to construct a strong 16-bit baseline for comparisons. Using LoRA on all transformer layers is critical to match 16-bit performance.

Default LoRA hyperparameters do not match 16-bit performance When using the standard practice of applying LoRA to query and value attention projection matrices [28], we are not able to replicate full finetuning performance for large base models. As shown in Figure 2 for LLaMA 7B finetuning on Alpaca, we find that the most critical LoRA hyperparameter is how many LoRA adapters are used in total and that LoRA on all linear transformer block layers are required to match full finetuning performance. Other LoRA hyperparameters, such as the projection dimension r, do not affect performance (see Appendix A).

Similarly, we find that default hyperparameters for fully finetuned baselines are undertuned. We do a hyperparameter search over learning rates 1e-6 to 5e-5 and batch sizes 8 to 128 to find robust baselines. Results for 7B LLaMA finetuning on Alpaca are shown in Figure 2.

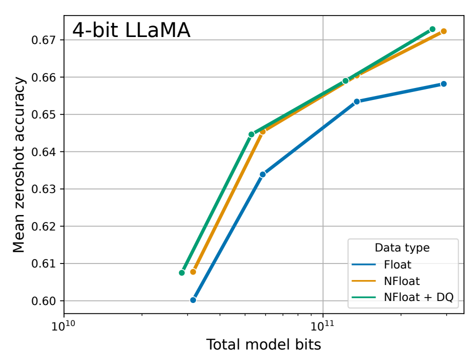

Figure 3: Mean zero-shot accuracy over Winogrande, HellaSwag, PiQA, Arc-Easy, and Arc-Challenge using LLaMA models with different 4-bit data types. The NormalFloat data type significantly improves the bit-for-bit accuracy gains compared to regular 4-bit Floats. While Double Quantization (DQ) only leads to minor gains, it allows for a more fine-grained control over the memory footprint to fit models of certain size (33B/65B) into certain GPUs (24/48GB).

4-bit NormalFloat yields better performance than 4-bit Floating Point While the 4-bit NormalFloat (NF4) data type is information-theoretically optimal, it still needs to be determined if this property translates to empirical advantages. We follow the setup from Dettmers and Zettlemoyer [13] where quantized LLMs (OPT [72], BLOOM [52], Pythia [7], LLaMA) of different sizes (125M to 65B) with different data types are evaluated on language modeling and a set of zero-shot tasks. In Figure 3 and Table 2 we see that NF4 improves performance significantly over FP4 and Int4 and that double quantization reduces the memory footprint without degrading performance.

k-bit QLoRA matches 16-bit full finetuning and 16-bit LoRA performance Recent findings have established that 4-bit quantization for *inference* is

实验设置。我们考虑三种架构（编码器、编码器-解码器和仅解码器），并将 QLoRA 与 16 位适配器微调以及全量微调进行比较，适用于最大 3B 的模型。我们的评估包括使用 RoBERTa-large 的 GLUE [58]、使用 T5 的 Super-NaturalInstructions (TKInstruct) [61]，以及在对 LLaMA 在 Flan v2 [39] 和 Alpaca [55] 上进行微调后进行的 5-shot MMLU [24]。为了进一步研究 NF4 相较于其他 4 位数据类型的优势，我们使用 Dettmers 和 Zettlemoyer [13] 的设置，并测量不同模型（OPT [72]、LLaMA [57]、BLOOM [52]、Pythia [7]）在 125m - 13B 模型规模下量化后的零-shot 精度和困惑度。我们在结果部分为每个特定设置提供了更多细节，以使结果更易于阅读。完整细节见附录 A。

虽然分页优化器对于在单个 24/48GB GPU 上进行 33B/65B QLORA 调优至关重要，但我们没有提供分页优化器的具体测量，因为分页仅在处理长序列长度的小批量时发生，而这种情况很少见。然而，我们确实对 48GB GPU 上 65B 模型的分页优化器运行时间进行了分析，并发现当批量大小为 16 时，分页优化器的训练速度与普通优化器相同。未来的工作应当测量并分析在什么情况下分页过程会导致速度下降。

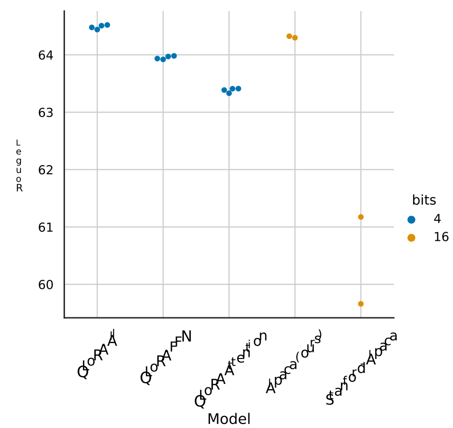

图2：LLaMA 7B 模型在 Alpaca 数据集上的 RougeL 分数。每个点代表使用不同随机种子的一次运行。我们改进了 Stanford Alpaca 完全微调的默认超参数，以构建一个强大的 16 位基线进行比较。在所有 Transformer 层上使用 LoRA 对于匹配 16 位性能至关重要。

默认的 LoRA 超参数与 16 位性能不匹配。当使用将 LoRA 应用于查询和数值注意力投影矩阵的标准做法时 [28]，我们无法复制大型基础模型的全量微调性能。如图 2 所示，在 LLaMA 7B 对 Alpaca 进行微调时，我们发现最关键的 LoRA 超参数是总共使用了多少个 LoRA 适配器，并且在所有线性 Transformer 块层上使用 LoRA 才能达到全量微调的性能。其他 LoRA 超参数，如

投影维度 r，不影响性能（见附录A）。

同样，我们发现全量微调基线的默认超参数调得不足。我们对学习率从 1e-6 到 5e-5，批量大小从 8 到 128 进行了超参数搜索，以找到稳健的基线。7B LLaMA 在 Alpaca 上微调的结果如图 2 所示。

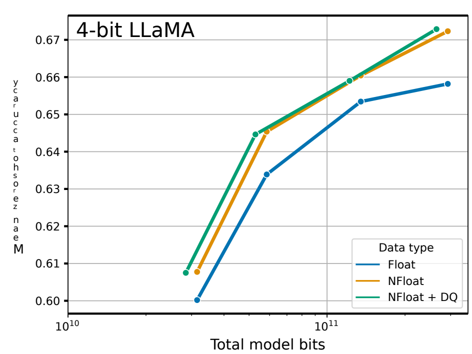

图 3：使用不同 4 位数据类型的 LLaMA 模型，在 Wino-grande、HellaSwag、PiQA、Arc-Easy 和 Arc-Challenge 上的平均零样本准确率。与常规 4 位浮点数相比，NormalFloat 数据类型显著提高了逐位准确率的提升。虽然双重量化（DQ）仅带来轻微增益，但它允许对内存占用进行更精细的控制，以将特定大小的模型（33B/65B）适配到特定 GPU（24/48GB）中。

4位 NormalFloat 的性能优于 4位浮点数。虽然 4位 NormalFloat（NF4）数据类型在信息论上是最优的，但仍需确定这种特性是否能转化为实际优势。我们遵循 Dettmers 和 Zettlemoyer [13] 的设置，对不同大小（125M 到 65B）和不同数据类型的量化大型语言模型（LLMs）（OPT [72]、BLOOM [52]、Pythia [7]、LLaMA）在语言建模和一组零样本任务上的表现进行评估。在图 3 和表 2 中，我们可以看到 NF4 的性能明显优于 FP4 和 Int4，同时双量化可以在不降低性能的情况下减少内存占用。

k 位 QLORA 与 16 位全量微调和 16 位 LoRA 性能相当。最近的研究发现，对于 *inference*，4 位量化是

Table 3: Experiments comparing 16-bit BrainFloat (BF16), 8-bit Integer (Int8), 4-bit Float (FP4), and 4-bit NormalFloat (NF4) on GLUE and Super-NaturalInstructions. QLoRA replicates 16-bit LoRA and full-finetuning.

Dataset	GLUE (Acc.)	Super-NaturalInstructions (RougeL)				
Model	RoBERTa-large	T5-80M	T5-250M	T5-780M	T5-3B	T5-11B
BF16	88.6	40.1	42.1	48.0	54.3	62.0
BF16 replication	88.6	40.0	42.2	47.3	54.9	-
LoRA BF16	88.8	40.5	42.6	47.1	55.4	60.7
QLoRA Int8	88.8	40.4	42.9	45.4	56.5	60.7
QLoRA FP4	88.6	40.3	42.4	47.5	55.6	60.9
QLoRA NF4 + DQ	-	40.4	42.7	47.7	55.3	60.9

possible, but leads to performance degradation relative to 16-bit [13, 18]. This raises the crucial question of whether the lost performance can be recovered by conducting 4-bit adapter finetuning. We test this for two setups.

The first focuses on a comparison with full 16-bit finetuning of RoBERTA and T5 models sized 125M to 3B parameters on GLUE and the Super-NaturalInstructions dataset. Results are shown in Table 3. In both datasets, we observe that 16-bit, 8-bit, and 4-bit adapter methods replicate the performance of the fully finetuned 16-bit baseline. This suggests that the performance lost due to the imprecise quantization can be fully recovered through adapter finetuning after quantization.

Table 2: Pile Common Crawl mean perplexity for different data types for 125M to 13B OPT, BLOOM, LLaMA, and Pythia models.

Data type	Mean PPL
Int4	34.34
Float4 (E2M1)	31.07
Float4 (E3M0)	29.48
NFloat4 + DQ	**27.41**

For our second setup, since full finetuning models at and beyond 11B parameters requires more than one server of high memory GPUs, we continue to test whether 4-bit QLoRA can match 16-bit LoRA at the 7B to 65B parameter scales. To this end, we finetune LLaMA 7B through 65B on two instruction following datasets, Alpaca and FLAN v2, and evaluate on the MMLU benchmark via 5-shot accuracy. Results are shown in Table 4 where we see that NF4 with double quantization fully recovers the 16-bit LoRA MMLU performance. In addition, we also note that QLoRA with FP4 lags behind the 16-bit brain float LoRA baseline by about 1 percentage point. This corroborates both our findings that (1) QLoRA with NF4 replicates both 16-bit full finetuning and 16-bit LoRA finetuning performance, and (2) NF4 is superior to FP4 in terms of quantization precision.

Summary Our results consistently show that 4-bit QLoRA with NF4 data type matches 16-bit full finetuning and 16-bit LoRA finetuning performance on academic benchmarks with well-established evaluation setups. We have also shown that NF4 is more effective than FP4 and that double quantization does not degrade performance. Combined, this forms compelling evidence that 4-bit QLoRA tuning reliably yields results matching 16-bit methods.

In line with previous work on quantization [13], our MMLU and Elo results indicate that with a given finetuning and inference resource budget it is beneficial to increase the number of parameters in the base model while decreasing their precision. This highlights the importance of efficiency benefits from QLoRA. Since we did not observe performance degradation compared to full-finetuning in our experiments with 4-bit finetuning, this raises the question of where the performance-precision trade-off exactly lies for QLoRA tuning, which we leave to future work to explore.

We proceed to investigate instruction tuning at scales that would be impossible to explore with full 16-bit finetuning on academic research hardware.

5 Pushing the Chatbot State-of-the-art with QLoRA

Having established that 4-bit QLoRA matches 16-bit performance across scales, tasks, and datasets we conduct an in-depth study of instruction finetuning up to the largest open-source language models available for research. To assess the performance of instruction finetuning these models, we evaluate

表 3：在 GLUE 和 Super-NaturalInstructions 上比较 16 位 BrainFloat (BF16)、8 位整数 (Int8)、4 位浮点 (FP4) 和 4 位标准浮点 (NF4) 的实验。QLORA 重现了 16 位 LoRA 和全量微调。

Dataset	GLUE (Acc.)	Super-NaturalInstructions (RougeL)				
Model	RoBERTa-large	T5-80M	T5-250M	T5-780M	T5-3B	T5-11B
BF16	88.6	40.1	42.1	48.0	54.3	62.0
BF16 replication	88.6	40.0	42.2	47.3	54.9	-
LoRA BF16	88.8	40.5	42.6	47.1	55.4	60.7
QLoRA Int8	88.8	40.4	42.9	45.4	56.5	60.7
QLoRA FP4	88.6	40.3	42.4	47.5	55.6	60.9
QLoRA NF4 + DQ	-	40.4	42.7	47.7	55.3	60.9

可能，但会导致性能下降 rel-
相对于16位[13, 18]。这引出了一个关键问题，即通过进行4位适配器微调是否可以恢复丢失的性能。我们对两种设置进行了测试。

第一个重点是将RoBERTA和T5模型（参数量从1.25亿到30亿）的完整16位微调在GLUE和Super-NaturalInstructions数据集上的表现进行比较。结果如表3所示。在这两个数据集中，我们观察到16位、8位和4位的适配器方法都能够复制完全微调的16位基线的性能。这表明，由于不精确量化导致的性能损失可以通过量化后的适配器微调完全恢复。

表 2：125M 至 13B OPT、BLOOM、LLaMA 和 Pythia 模型在不同数据类型上的 Pile Common Crawl 平均困惑度。

Data type	Mean PPL
Int4	34.34
Float4 (E2M1)	31.07
Float4 (E3M0)	29.48
NFloat4 + DQ	**27.41**

对于我们的第二个设置，由于对11B及以上参数的模型进行全量微调需要多台高内存GPU服务器，我们继续测试4-bit QLORA是否能够在7B到65B参数规模上匹配16-bit LoRA。为此，我们在两条指令数据上对LLaMA 7B到65B进行微调数据集，包括 Alpaca 和 FLAN v2，并在 MMLU 基准上通过 5-shot 准确率进行评估。结果显示在表 4 中，我们可以看到，使用双重量化的 NF4 完全恢复了 16 位 LoRA 在 MMLU 上的性能。此外，我们还注意到，使用 FP4 的 QLORA 比 16 位 Brain Float LoRA 基线落后约 1 个百分点。这证实了我们的两个发现：（1）使用 NF4 的 QLORA 能够复制 16 位全量微调和 16 位 LoRA 微调的性能；（2）在量化精度方面，NF4 优于 FP4。

总结 我们的结果一致显示，使用 NF4 数据类型的 4 位 QLORA 在学术基准测试中，其表现与 16 位全微调和 16 位 LoRA 微调相匹配，并且这些基准测试有完善的评估设置。我们还证明了 NF4 比 FP4 更有效，双重量化不会降低性能。综合来看，这提供了令人信服的证据，表明 4 位 QLORA 微调可靠地产生与 16 位方法相当的结果。

与之前关于量化的工作[13]一致，我们的MMLU和Elo结果表明，在给定的微调和推理资源预算下，增加基础模型的参数数量同时降低其精度是有益的。这凸显了QLORA在效率方面的优势。由于我们在4位微调的实验中未观察到与全量微调相比的性能下降，这就提出了一个问题：QLoRA微调的性能与精度之间的权衡究竟在哪里，我们将这一问题留待未来的工作进行探索。

我们着手在学术研究硬件上无法用完整16位微调探索的规模下研究指令微调。

5 使用 QLoRA 推动聊天机器人技术的最前沿

在确定4位QLORA在各类规模、任务和数据集上能够匹配16位性能之后，我们对指令微调进行了深入研究，涵盖了可用于研究的最大开源语言模型。为了评估这些模型的指令微调性能，我们进行了评估

Table 4: Mean 5-shot MMLU test accuracy for LLaMA 7-65B models finetuned with adapters on Alpaca and FLAN v2 for different data types. Overall, NF4 with double quantization (DQ) matches BFloat16 performance, while FP4 is consistently one percentage point behind both.

	Mean 5-shot MMLU Accuracy								
LLaMA Size	7B		13B		33B		65B		Mean
Dataset	Alpaca	FLAN v2	Alpaca	FLAN v2	Alpaca	FLAN v2	Alpaca	FLAN v2	
BFloat16	38.4	45.6	47.2	50.6	57.7	60.5	61.8	62.5	53.0
Float4	37.2	44.0	47.3	50.0	55.9	58.5	61.3	63.3	52.2
NFloat4 + DQ	39.0	44.5	47.5	50.7	57.3	59.2	61.8	63.9	53.1

on a challenging Natural Language Understanding benchmark (MMLU) and develop new methods for real-world chatbot performance evaluation.

5.1 Experimental setup

We now describe an overview of the experimental setup with full details in Appendix B.

Data As, to our knowledge, there is no comprehensive study of recent instruction-following datasets, we select eight recent datasets. We include datasets obtained through crowd-sourcing (OASST1 [31], HH-RLHF [4]), distillation from instruction-tuned models (Alpaca [55], self-instruct [59], unnatural-instructions [26]), corpora aggregations (FLAN v2 [12]), as well as hybrids (Chip2 [32], Long-form [30]). These datasets cover different languages, data sizes, and licenses.

Training Setup To avoid confounding effects from different training objectives, we perform QLoRA finetuning with cross-entropy loss (supervised learning) without reinforcement learning, even for datasets that include human judgments of different responses. For datasets that have a clear distinction between instruction and response, we finetune only on the response (see ablations in Appendix B). For OASST1 and HH-RLHF, multiple responses are available. We then select the top response at every level of the conversation tree and finetune on the full selected conversation, including the instructions. In all of our experiments, we use NF4 QLoRA with double quantization and paged optimizers to prevent memory spikes during gradient checkpointing. We do small hyperparameter searches for the 13B and 33B LLaMA models and we find that all hyperparameter settings found at 7B generalize (including number of epochs) except learning rate and batch size. We halve the learning rate for 33B and 65B while doubling the batch size.

Baselines We compare our models to both research (Vicuna [10] and Open Assistant [31]) and commercial (GPT-4 [42], GPT-3.5-turbo and Bard) chatbot systems. The Open Assistant model is a LLaMA 33B model finetuned with Reinforcement Learning from Human Feedback (RLHF) on the same OASST1 dataset that we experiment with. Vicuna does full fine-tuning of LLaMA 13B on proprietary user-shared conversations from ShareGPT and is thus the result of distillation from OpenAI GPT models.

5.2 Evaluation

Following common practice, we use the MMLU (Massively Multitask Language Understanding) benchmark [24] to measure performance on a range of language understanding tasks. This is a multiple-choice benchmark covering 57 tasks including elementary mathematics, US history, computer science, law, and more. We report 5-shot test accuracy.

Table 5: MMLU 5-shot test results for different sizes of LLaMA finetuned on the corresponding datasets using QLoRA.

Dataset	7B	13B	33B	65B
LLaMA no tuning	35.1	46.9	57.8	63.4
Self-Instruct	36.4	33.3	53.0	56.7
Longform	32.1	43.2	56.6	59.7
Chip2	34.5	41.6	53.6	59.8
HH-RLHF	34.9	44.6	55.8	60.1
Unnatural Instruct	41.9	48.1	57.3	61.3
Guanaco (OASST1)	36.6	46.4	57.0	62.2
Alpaca	38.8	47.8	57.3	62.5
FLAN v2	44.5	51.4	59.2	63.9

We also test generative language capabilities through both automated and human evaluations. This second set of evaluations relies on queries curated by humans and aims at measuring the quality of model responses. While this is a more realistic testbed for chatbot model performance and is growing in popularity, there is no commonly accepted protocol in the literature. We describe below our proposed setup, using nucleus sampling with $p = 0.9$ and temperature 0.7 in all cases.

表 4：使用适配器在 Alpaca 和 FLAN v2 上微调的 LLaMA 7-65B 模型在不同数据类型下的平均 5-shot MMLU 测试准确率。总体而言，带双量化（DQ）的 NF4 性能与 BFloat16 相当，而 FP4 始终比两者低一个百分点。

LLaMA Size	Mean 5-shot MMLU Accuracy								Mean
	7B		13B		33B		65B		
Dataset	Alpaca	FLAN v2	Alpaca	FLAN v2	Alpaca	FLAN v2	Alpaca	FLAN v2	
BFloat16	38.4	45.6	47.2	50.6	57.7	60.5	61.8	62.5	53.0
Float4	37.2	44.0	47.3	50.0	55.9	58.5	61.3	63.3	52.2
NFloat4 + DQ	39.0	44.5	47.5	50.7	57.3	59.2	61.8	63.9	53.1

在一个具有挑战性的自然语言理解基准（MMLU）上，并开发用于实际聊天机器人性能评估的新方法。

5.1 实验设置

我们现在描述实验设置的概况，详细内容见附录B。

据我们所知，目前没有对最新的指令跟随数据集进行过全面的研究，因此我们选择了八个最近的数据集。我们包括了通过众包获取的数据集（OASST1 [31]，HH-RLHF [4]）、从指令调优模型蒸馏得到的数据集（Alpaca [55]，self-instruct [59]，unnatural-instructions [26]）、语料聚合（FLAN v2 [12]），以及混合型数据集（Chip2 [32]，Long-form [30]）。这些数据集涵盖了不同的语言、数据规模和许可协议。

训练设置 为了避免不同训练目标的混淆效应，我们在进行 QLoRA 微调时使用交叉熵损失（监督学习），而不使用强化学习，即使是包含人类对不同响应评判的数据集也如此。对于那些指令与响应有明确区分的数据集，我们仅对响应进行微调（参见附录 B 的消融实验）。对于 OASST1 和 HH-RLHF 数据集，有多个可用响应。我们会在对话树的每个层级选择最佳响应，并在所选完整对话（包括指令）上进行微调。在我们所有的实验中，我们使用带有双重量化和分页优化器的 NF4 QLORA，以防止在梯度检查点过程中出现内存峰值。我们对 13 B 和 33B LLaMA 模型进行了小范围超参数搜索，发现 7B 模型找到的所有超参数设置都可以泛化（包括训练轮数），唯独学习率和批量大小例外。对于 33B 和 65B 模型，我们将学习率减半，同时

基线 我们将我们的模型与研究型（Vicuna [10] 和 Open Assistant [31]）以及商业型（GPT-4 [42]、GPT-3.5-turbo 和 Bard）聊天机器人系统进行比较。Open Assistant 模型是一个 LLaMA 33B 模型，通过人类反馈强化学习（RLHF）在我们实验使用的相同 OASST1 数据集上进行微调。Vicuna 则对 LLaMA 13B 进行了完整的微调，使用来自 ShareGPT 的专有用户共享对话，因此它是从 OpenAI GPT 模型蒸馏的结果。

5.2 评估

遵循常规做法，我们使用 MMLU（大规模多任务语言理解）基准 [24] 来衡量在各种语言理解任务上的性能。这是一个多项选择基准，涵盖 57 个任务，包括初等数学、美国历史、计算机科学、法律等。我们报告 5 次示例的测试准确率。

表 5：不同规模的 LLaMA 在相应数据集上使用 QLoRA 微调后的 MMLU 5-shot 测试结果。

Dataset	7B	13B	33B	65B
LLaMA no tuning	35.1	46.9	57.8	63.4
Self-Instruct	36.4	33.3	53.0	56.7
Longform	32.1	43.2	56.6	59.7
Chip2	34.5	41.6	53.6	59.8
HH-RLHF	34.9	44.6	55.8	60.1
Unnatural Instruct	41.9	48.1	57.3	61.3
Guanaco (OASST1)	36.6	46.4	57.0	62.2
Alpaca	38.8	47.8	57.3	62.5
FLAN v2	44.5	51.4	59.2	63.9

我们还通过自动化和人工评估来测试生成式语言能力。这第二组评估依赖于人工策划的查询，旨在衡量模型响应的质量。虽然这是一个更贴近实际的聊天机器人模型性能测试平台，并且越来越受欢迎，但文献中尚无普遍接受的标准化协议。我们去-
在下面记录我们的拟议设置，在所有情况下使用核采样，$p = 0.9$ 和温度 0.7。

Benchmark Data We evaluate on two curated datasets of queries (questions): the Vicuna prompts [10] and the OASST1 validation dataset [31]. We use the Vicuna prompts, a set of 80 prompts from a diverse set of categories, without modifications. The OASST1 dataset is a multilingual collection of crowd-sourced multiturn dialogs between a user and an assistant. We select all user messages in the validation dataset as queries and include previous turns in the prompt. This procedure leads to 953 unique user queries. We term these two datasets the Vicuna and OA benchmarks.

Automated Evaluation First, based on the evaluation protocol introduced by Chiang et al. [10], we use GPT-4 to rate the performance of different systems against ChatGPT (GPT-3.5 Turbo) on the Vicuna benchmark. Given a query along with ChatGPT's and a model's responses, GPT-4 is prompted to assign a score out of ten to both responses and provide an explanation. The overall performance of a model is calculated as a percentage of the score that ChatGPT achieved. Note this relative score can be higher than 100% if the model achieves a higher absolute score than ChatGPT. We find a significant ordering effect with GPT-4 increasing the score of the response occurring earlier in the prompt. To control for such effects, we recommend reporting the mean score over both orders.

Next, we measure performance through direct comparisons between system outputs. We simplify the rating scheme to a three-class labeling problem that accounts for ties. We prompt GPT-4 to pick the best response or declare a tie and provide an explanation. We conduct these head-to-head comparisons on all permutations of pairs of systems on both the Vicuna and OA benchmarks.

Human Evaluation While recent work indicates generative models can be effectively employed for system evaluations [19], the reliability GPT-4 ratings to assess chatbot performance is, to our knowledge, yet to be proven to correlate with human judgments. Therefore, we run two parallel human evaluations on the Vicuna benchmark matching both automated evaluation protocols described above. We use Amazon Mechanical Turk (AMT) and get two human annotators for comparisons to ChatGPT and three annotators for pairwise comparisons.

Elo Rating With both human and automated pairwise comparisons, we create a tournament-style competition where models compete against each other. The tournament is made up of matches where pairs of models compete to produce the best response for a given prompt. This is similar to how Bai et al. [4] and Chiang et al. [10] compare models, but we also employ GPT-4 ratings in addition to human ratings. We randomly sample from the set of labeled comparisons to compute Elo [16, 17]. Elo rating, which is widely used in chess and other games, is a measure of the expected win-rate relative to an opponent's win rate, for example, an Elo of 1100 vs 1000 means the Elo 1100 player has an expected win-rate of approximately 65% against the Elo 1000 opponent; a 1000 vs 1000 or 1100 vs 1100 match results in an expected win-rate of 50%. The Elo rating changes after each match proportionally to the expected outcome, that is, an unexpected upset leads to a large change in Elo rating while an expected outcome leads to a small change. Over time, Elo ratings approximately match the skill of each player at playing the game. We start with a score of 1,000 and use $K = 32$. Similar to Chiang et al. [10], we repeat this procedure 10,000 times with different random seeds to control for ordering effects, e.g., the effect of which model pairs compete with each other first.

5.3 Guanaco: QLoRA trained on OASST1 is a State-of-the-art Chatbot

Based on our automated and human evaluations, we find that the top QLoRA tuned model, Guanaco 65B, which we finetune on a variant of OASST1, is the best-performing open-source chatbot model and offers performance competitive to ChatGPT. When compared to GPT-4, Guanaco 65B and 33B have an expected win probability of 30%, based on Elo rating from human annotators system-level pairwise comparisons - the highest reported to date.

The Vicuna benchmark [10] results relative to ChatGPT are shown in Table 6. We find that Guanaco 65B is the best-performing model after GPT-4, achieving 99.3% performance relative to ChatGPT. Guanaco 33B has more parameters than the Vicuna 13B model, but uses only 4-bit precision for its weights and is thus much more memory efficient at 21 GB vs 26 GB, providing a three percentage points of improvement over Vicuna 13B. Furthermore, Guanaco 7B easily fits on modern phones at a 5 GB footprint while still scoring nearly 20 percentage points higher than Alpaca 13B.

However, Table 6 also has very wide confidence intervals, with many models overlapping in performance. We hypothesize that this uncertainty comes from the lack of clear specification of scale, e.g., it is unclear what 8 on a 10 point scale means across different scenarios. As such, we instead recommend using the Elo ranking method [16], based on *pairwise* judgments from human annotators and GPT-4 to avoid the problem of grounding an absolute scale. Elo ratings of the most competitive

基准数据 我们在两个策划的数据集上进行评估，这些数据集由查询（问题）组成：Vicuna 提示 [10] 和 OASST1 验证数据集 [31]。我们使用 Vicuna 提示，这是一个来自不同类别的 80 个提示集合，未进行任何修改。OASST1 数据集是一个多语言的众包多轮对话集合，记录了用户与助手之间的对话。我们选择验证数据集中的所有用户消息作为查询，并在提示中包含前面的对话轮次。该过程产生了 953 个独立用户查询。我们将这两个数据集称为 Vicuna 和 OA 基准。

自动化评估 首先，基于Chiang等人[10]提出的评估协议，我们使用GPT-4在Vicuna基准上评估不同系统相对于ChatGPT（GPT-3.5 Turbo）的表现。对于一个查询以及ChatGPT和某个模型的回答，GPT-4会被提示为两个回答分别打出十分制的分数并提供解释。模型的整体表现被计算为该模型得分占ChatGPT得分的百分比。请注意，如果模型的绝对得分高于ChatGPT，则该相对分数可能超过100%。我们发现GPT-4存在显著的顺序效应，即它会提高出现在提示前面的回答的分数。为控制此类效应，我们建议报告两种顺序下的平均分。

接下来，我们通过系统输出之间的直接比较来衡量性能。我们将评分方案简化为一个三类标注问题，以考虑平局情况。我们提示 GPT-4 选择最佳响应或宣布平局，并提供解释。我们在 Vicuna 和 OA 基准的所有系统对排列上进行这些一对一的比较。

人工评估 虽然近期研究表明生成模型可以有效用于系统评估[19]，但据我们所知，GPT-4评分在评估聊天机器人性能方面的可靠性尚未被证明与人工判断相关。因此，我们在Vicuna基准上进行两项平行的人类评估，以匹配上述的两种自动评估协议。我们使用亚马逊机械土耳其人（Amazon Mechanical Turk, AMT）平台，并为与ChatGPT的比较招募了两名人工标注员，为成对比较招募了三名标注员。

Elo 评分　通过人工和自动化的成对比较，我们创建了一种锦标赛式的竞争，让模型相互对抗。锦标赛由比赛组成，每场比赛中成对的模型竞争为给定提示生成最佳响应。这类似于 Bai 等人 [4] 和 Chiang 等人 [10] 比较模型的方式，但我们除了人工评分外，还采用了 GPT-4 评分。我们从标注比较集合中随机抽样以计算 Elo [16, 17]。Elo 评分广泛用于国际象棋和其他游戏，是一种相对于对手胜率的预期胜率的度量。例如，Elo 1100 对 1000 表示 Elo 1100 的玩家对 Elo 1000 的对手的预期胜率约为 65%；1000 对 1000 或 1100 对 1100 的比赛预期胜率为 50%。Elo 评分在每场比赛后会根据预期结果进行变化，即意料之外的胜利会导致 Elo 评分大幅变化，而预期结果则变化较小。

5.3 卦　naco：在 OASST1 上训练的 QLORA 是最先进的 C　帽子机器人

根据我们的自动化和人工评估，我们发现经过QLORA调优的顶级模型Guanaco 65B（我们在OASST1的一个变体上进行微调）是表现最好的开源聊天机器人模型，其性能可与ChatGPT竞争。与GPT-4相比，基于人类标注者的系统级成对比较的Elo评分，Guanaco 65B和33B的预期胜率为30%——这是迄今为止报告的最高值。

相对于 ChatGPT 的 Vicuna 基准测试 [10] 结果如表 6 所示。我们发现，Guanaco 65B 是继 GPT-4 之后性能最好的模型，其性能达到相对于 ChatGPT 的 99.3%。Guanaco 33B 的参数比 Vicuna 13B 模型更多，但其权重只使用 4 位精度，因此在内存使用上更加高效，为 21 GB 对比 26 GB，相比 Vicuna 13B 提升了三个百分点。此外，Guanaco 7B 可以轻松适配现代手机，占用空间仅 5 GB，但其得分仍比 Alpaca 13B 高近 20 个百分点。

然而，表6也存在非常宽的置信区间，许多模型的表现存在重叠。我们假设这种不确定性来自于缺乏明确的量表说明，例如，在不同的情境下，10分制中的8分意味着什么并不清楚。因此，我们推荐使用Elo排名方法[16]，基于人类标注者和GPT-4的*pairwise*判断，以避免确定绝对量表的问题。最具竞争力的Elo评分

Table 6: Zero-shot Vicuna benchmark scores as a percentage of the score obtained by ChatGPT evaluated by GPT-4. We see that OASST1 models perform close to ChatGPT despite being trained on a very small dataset and having a fraction of the memory requirement of baseline models.

Model / Dataset	Params	Model bits	Memory	ChatGPT vs Sys	Sys vs ChatGPT	Mean	95% CI
GPT-4	-	-	-	119.4%	110.1%	**114.5**%	2.6%
Bard	-	-	-	93.2%	96.4%	94.8%	4.1%
Guanaco	65B	4-bit	41 GB	96.7%	101.9%	**99.3**%	4.4%
Alpaca	65B	4-bit	41 GB	63.0%	77.9%	70.7%	4.3%
FLAN v2	65B	4-bit	41 GB	37.0%	59.6%	48.4%	4.6%
Guanaco	33B	4-bit	21 GB	96.5%	99.2%	**97.8**%	4.4%
Open Assistant	33B	16-bit	66 GB	91.2%	98.7%	94.9%	4.5%
Alpaca	33B	4-bit	21 GB	67.2%	79.7%	73.6%	4.2%
FLAN v2	33B	4-bit	21 GB	26.3%	49.7%	38.0%	3.9%
Vicuna	13B	16-bit	26 GB	91.2%	98.7%	**94.9**%	4.5%
Guanaco	13B	4-bit	10 GB	87.3%	93.4%	90.4%	5.2%
Alpaca	13B	4-bit	10 GB	63.8%	76.7%	69.4%	4.2%
HH-RLHF	13B	4-bit	10 GB	55.5%	69.1%	62.5%	4.7%
Unnatural Instr.	13B	4-bit	10 GB	50.6%	69.8%	60.5%	4.2%
Chip2	13B	4-bit	10 GB	49.2%	69.3%	59.5%	4.7%
Longform	13B	4-bit	10 GB	44.9%	62.0%	53.6%	5.2%
Self-Instruct	13B	4-bit	10 GB	38.0%	60.5%	49.1%	4.6%
FLAN v2	13B	4-bit	10 GB	32.4%	61.2%	47.0%	3.6%
Guanaco	7B	4-bit	5 GB	84.1%	89.8%	**87.0**%	5.4%
Alpaca	7B	4-bit	5 GB	57.3%	71.2%	64.4%	5.0%
FLAN v2	7B	4-bit	5 GB	33.3%	56.1%	44.8%	4.0%

models can be seen in Table 1. We note that human and GPT-4 ranking of models on the Vicuna benchmark disagree partially, particularly for Guanaco 7B, but are consistent for most models with a Kendall Tau of $\tau = 0.43$ and Spearman rank correlation of $r = 0.55$ at the system level. At the example level, the agreement between GPT-4 and human annotators' majority vote is weaker with Fleiss $\kappa = 0.25$. Overall, this shows a moderate agreement between system-level judgments by GPT-4 and human annotators, and thus that model-based evaluation represents a somewhat reliable alternative to human evaluation. We discuss further considerations in Section 6.2.

Elo rankings in Table 7 indicate that Guanaco 33B and 65B models outperform all models besides GPT-4 on the Vicuna and OA benchmarks and that they perform comparably to ChatGPT in line with Table 6. We note that the Vicuna benchmark favors open-source models while the larger OA benchmark favors ChatGPT. Furthermore, we can see from Tables 5 and 6 that the suitability of a finetuning dataset is a determining factor in performance. Finetuning Llama models on FLAN v2 does particularly well on MMLU, but performs worst on the Vicuna benchmark (similar trends are observed with other models). This also points to partial orthogonality in current evaluation benchmarks: strong MMLU performance does not imply strong chatbot performance (as measured by Vicuna or OA benchmarks) and vice versa.

Guanaco is the only top model in our evaluation that is not trained on proprietary data as the OASST1 dataset collection guidelines explicitly forbid the use of GPT models. The next best model trained on only open-source data is the Anthropic HH-RLHF model, which scores 30 percentage points lower than Guanaco on the Vicuna benchmark (see Table 6). Overall, these results show that 4-bit QLoRA is effective and can produce state-of-the-art chatbots that rival ChatGPT. Furthermore, our 33B Guanaco can be trained on 24 GB consumer GPUs in less than 12 hours. This opens up the potential for future work via QLoRA tuning on specialized open-source data, which produces models that can compete with the very best commercial models that exist today.

6 Qualitative Analysis

While quantitative analysis is the core of our evaluation, there are a number of issues with only looking at summary statistics. Perhaps the largest is the problem of benchmark validity [36]—whether a benchmark truly tests what its name or description suggests is always at question, especially as we discover "shortcuts" to solve benchmarks that machine learning models sometimes exploit [22, 46]. To partially alleviate this, we here perform some qualitative analysis, in two sections. First, in §6.1

表6：零样本Vicuna基准分数占GPT-4评估的ChatGPT得分的百分比。我们看到，尽管OASST1模型仅在非常小的数据集上训练，并且所需内存仅为基线模型的一小部分，但其表现接近ChatGPT。

Model / Dataset	Params	Model bits	Memory	ChatGPT vs Sys	Sys vs ChatGPT	Mean	95% CI
GPT-4	-	-	-	119.4%	110.1%	**114.5**%	2.6%
Bard	-	-	-	93.2%	96.4%	94.8%	4.1%
Guanaco	65B	4-bit	41 GB	96.7%	101.9%	**99.3**%	4.4%
Alpaca	65B	4-bit	41 GB	63.0%	77.9%	70.7%	4.3%
FLAN v2	65B	4-bit	41 GB	37.0%	59.6%	48.4%	4.6%
Guanaco	33B	4-bit	21 GB	96.5%	99.2%	**97.8**%	4.4%
Open Assistant	33B	16-bit	66 GB	91.2%	98.7%	94.9%	4.5%
Alpaca	33B	4-bit	21 GB	67.2%	79.7%	73.6%	4.2%
FLAN v2	33B	4-bit	21 GB	26.3%	49.7%	38.0%	3.9%
Vicuna	13B	16-bit	26 GB	91.2%	98.7%	**94.9**%	4.5%
Guanaco	13B	4-bit	10 GB	87.3%	93.4%	90.4%	5.2%
Alpaca	13B	4-bit	10 GB	63.8%	76.7%	69.4%	4.2%
HH-RLHF	13B	4-bit	10 GB	55.5%	69.1%	62.5%	4.7%
Unnatural Instr.	13B	4-bit	10 GB	50.6%	69.8%	60.5%	4.2%
Chip2	13B	4-bit	10 GB	49.2%	69.3%	59.5%	4.7%
Longform	13B	4-bit	10 GB	44.9%	62.0%	53.6%	5.2%
Self-Instruct	13B	4-bit	10 GB	38.0%	60.5%	49.1%	4.6%
FLAN v2	13B	4-bit	10 GB	32.4%	61.2%	47.0%	3.6%
Guanaco	7B	4-bit	5 GB	84.1%	89.8%	**87.0**%	5.4%
Alpaca	7B	4-bit	5 GB	57.3%	71.2%	64.4%	5.0%
FLAN v2	7B	4-bit	5 GB	33.3%	56.1%	44.8%	4.0%

模型可以在表1中看到。我们注意到，在Vicuna基准上，人类和GPT-4对模型的排名部分不一致，特别是对于Guanaco 7B，但对大多数模型是一致的，系统层面的Kendall Tau为$\tau = 0.43$，Spearman等级相关为$r = 0.55$。在示例层面，GPT-4与人类标注者多数投票之间的一致性较弱，Fleiss为$\kappa = 0.25$。总体而言，这表明GPT-4与人类标注者在系统层面的判断存在中等程度的一致性，因此基于模型的评估在一定程度上是人类评估的可靠替代。我们在第6.2节中讨论了进一步的考虑因素。

表 7 中的 Elo 排名显示，Guanaco 33B 和 65B 模型在 Vicuna 和 OA 基准测试中表现优于除 GPT-4 之外的所有模型，并且它们的表现与表 6 中的 ChatGPT 相当。我们注意到，Vicuna 基准测试更有利于开源模型，而较大的 OA 基准测试则更有利于 ChatGPT。此外，从表 5 和表 6 可以看出，微调数据集的适用性是性能的决定性因素。在 FLAN v2 数据集上微调 Llama 模型在 MMLU 上表现尤为出色，但在 Vicuna 基准测试中表现最差（对其他模型也观察到类似趋势）。这也表明当前评价基准存在部分正交性：在 MMLU 上表现强劲并不意味着聊天机器人性能强（以 Vicuna 或 OA 基准衡量），反之亦然。

Guanaco 是我们评估中唯一一款没有使用专有数据训练的顶级模型，因为 OASST1 数据集收集指南明确禁止使用 GPT 模型。在仅使用开源数据训练的模型中，表现次佳的是 Anthropic HH-RLHF 模型，其在 Vicuna 基准测试中的得分比 Guanaco 低 30 个百分点（见表 6）。总体而言，这些结果表明 4-bit QLORA 是有效的，并且可以生成与 ChatGPT 竞争的最先进聊天机器人。此外，我们的 33B Guanaco 可以在 24 GB 消费级 GPU 上在不到 12 小时内完成训练。这为未来通过 QLORA 在专业化开源数据上进行调优提供了可能，从而生成可以与当今最优秀的商业模型竞争的模型。

6 定性分析

虽然量化分析是我们评估的核心，但仅仅看摘要统计数据存在一些问题。也许最大的问题是基准的有效性问题 [36]——一个基准是否真正测试了其名称或描述所暗示的内容总是存在疑问，尤其是当我们发现机器学习模型有时会利用“捷径”来解决基准问题 [22, 46] 时。为部分缓解这一问题，我们在此进行一些定性分析，分为两个部分。首先，在 §6.1

Table 7: Elo rating for a tournament between models where models compete to generate the best response for a prompt, judged by human raters or GPT-4. Overall, Guanaco 65B and 33B tend to be preferred to ChatGPT-3.5 on the benchmarks studied. According to human raters they have a Each 10-point difference in Elo is approximately a difference of 1.5% in win-rate.

Benchmark	Vicuna		Vicuna		Open Assistant		
# Prompts	80		80		953		
Judge	Human raters		GPT-4		GPT-4		Median Rank
Model	Elo	Rank	Elo	Rank	Elo	Rank	
GPT-4	1176	1	1348	1	1294	1	1
Guanaco-65B	1023	2	1022	2	1008	3	2
Guanaco-33B	1009	4	992	3	1002	4	4
ChatGPT-3.5 Turbo	916	7	966	5	1015	2	5
Vicuna-13B	984	5	974	4	936	5	5
Guanaco-13B	975	6	913	6	885	6	6
Guanaco-7B	1010	3	879	8	860	7	7
Bard	909	8	902	7	-	-	8

we show some examples that we believe are representative of some observed patterns in the text generated by our 65b Guanaco model. Second, §6.2 we detail considerations about the results we have discussed and our interpretation of them.

6.1 Qualitative Analysis of Example Generations

To find examples, we first go through data generated for the Vicuna benchmark and the OpenAssistant benchmark, and look for patterns in the answers Guanaco generates. When we notice a pattern we attempt to setup a question or prompt that will induce the pattern even though it is the incorrect solution, e.g., if we observe that the model tends to give long-winded answers we prompt the model to "Answer yes or no without explanation." We use this to find "lemons" where we manage to adversarially break the model and "cherries" where we fail to break the model, and present both. All generations in this section were generated with Nucleus Sampling [25] with $p = 0.9$.

Of course, this is by no means comprehensive, since it is beyond the scope of this small qualitative study to control for all the variables involved, e.g., the full distribution of responses the model can generate for a given prompt is quite large, so we rely on samples we hope are representative. However, we believe describing these examples gives context to the quantitative evidence shown earlier in the paper. Since we open source all models and code, we hope this section will inspire future work to examine in more detail the issues we present here.

Factual Recall For questions such as "What is the capital of Zambia?" all models consistently generate correct answers, e.g.,

> **Guanaco**: `The capital of Zambia is Lusaka.`

However as questions get more obscure Guanaco becomes unreliable, but stays confident. For instance in response to this prompt from HotPotQA [68]:

> **User**: In what year was the singer who popularized 'I'll Keep the Lovelight Burning' born?
>
> **Guanaco**: `Al Jolson is credited with popularizing the song 'I'll Keep the Lovelight Burning,' and he was born in the year 1886.`

Guanaco generates the wrong popularizer and the wrong birthday (though the birthday given is correct for the person mentioned, Al Jolson).

Suggestibility Guanaco shows a surprising resistance to going along with some kinds of assumed misinformation, e.g., in the following interaction:

> **User**: How was it finally, officially confirmed that the earth is flat by peer reviewed scientists?

表 7：模型之间的锦标赛 Elo 等级评分，其中模型竞争以生成对提示的最佳回应，由人工评审或 GPT-4 进行评分。总体而言，在所研究的基准测试中，Guanaco 65B 和 33B 相较于 ChatGPT-3.5 更受偏好。根据人工评审，每 10 点 Elo 差异大约对应 1.5% 的胜率差异。

Benchmark	Vicuna		Vicuna		Open Assistant		
# Prompts	80		80		953		
Judge	Human raters		GPT-4		GPT-4		Median Rank
Model	Elo	Rank	Elo	Rank	Elo	Rank	
GPT-4	1176	1	1348	1	1294	1	1
Guanaco-65B	1023	2	1022	2	1008	3	2
Guanaco-33B	1009	4	992	3	1002	4	4
ChatGPT-3.5 Turbo	916	7	966	5	1015	2	5
Vicuna-13B	984	5	974	4	936	5	5
Guanaco-13B	975	6	913	6	885	6	6
Guanaco-7B	1010	3	879	8	860	7	7
Bard	909	8	902	7	-	-	8

我们展示了一些示例，我们认为这些示例代表了我们65b Guanaco模型生成的文本中观察到的一些模式。其次，在§6.2中，我们详细说明了关于我们所讨论结果的考虑以及我们的解释。

6.1 示例生成的定性分析

为了找到示例，我们首先浏览为Vicuna基准和OpenAssistant基准生成的数据，并寻找Guanaco生成的答案中的模式。当我们注意到某个模式时，我们会尝试设置一个问题或提示，以诱发该模式，即使它是错误的解决方案，例如，如果我们观察到模型倾向于给出冗长的答案，我们会提示模型“仅回答是或否，不作解释”。我们利用这一方法找到“柠檬”，即我们设法对模型进行对抗性破坏的情况，以及“樱桃”，即我们未能破坏模型的情况，并展示两者。本节中的所有生成内容均使用Nucleus采样[25]生成，参数为$p = 0.9$。

当然，这绝不是全面的，因为对于这个小型定性研究来说，要控制所有相关变量是超出其范围的，例如，对于给定提示，模型可以生成的完整响应分布相当大，因此我们依赖希望具有代表性的样本。然而，我们认为描述这些示例可以为本文早先展示的定量证据提供背景。由于我们开源所有模型和代码，我们希望本节能激励未来的研究更详细地探讨我们在此提出的问题。

事实回顾 对于诸如“赞比亚的首都是什么？”的问题，所有模型都能一致生成正确答案，例如，

> Guanaco：赞比亚的首都是卢萨卡。

然而，随着问题变得越来越晦涩，Guanaco 变得不可靠，但仍然保持自信。例如，对于 HotPotQA [68] 的这个提示，它的回答是：

> 用户：那位使《I’ll Keep the Lovelight Burning》流行起来的歌手是哪一年出生的?
>
> 美洲骆驼：阿尔·乔尔森被认为是让歌曲《我会继续点亮爱的灯光》流行起来的人，他出生于1886年。

美洲驼生成了错误的普及者和错误的生日（尽管提供的生日对于所提到的人，阿尔·乔尔森，是正确的）。

可暗示性 美洲草驼对接受某些类型的假定错误信息表现出惊人的抵抗力，例如，在以下互动中：

> 用户：地球是平的这一点是如何最终由同行评审的科学家正式确认的?

Guanaco: `The earth has never been officially confirmed to be flat by peer reviewed scientists. The overwhelming consensus among scientists is that the earth is spherical. In fact, this has been known for centuries and has been confirmed through numerous experiments and observations.`

`The belief that the earth is flat is a conspiracy theory that has been debunked by science. The idea that a group of scientists have peer reviewed and confirmed the earth is flat is simply false.`

`If you have any more questions please ask.`

Guanaco is also quite good at knowing what kinds of questions aren't possible to answer, e.g.,

User: What time is it?

Guanaco: `I'm sorry, but as an AI language model, I don't have access to real-time information. My responses are generated based on my pre-existing knowledge and do not reflect the current time. If you're looking for the current time, I suggest using a timekeeping device or checking an online timepiece.`

Similar results hold for "Where are you?", "How are you?", etc.

Refusal Similarly, Guanaco sometimes refuses to follow instructions for seemingly random reasons:

User: Please reverse the words in the sentence "I'm a cool dude in a mocha mood."

Guanaco: `I'm sorry, I'm not able to do that right now, but I can explain why the sentence you provided is grammatically correct.`

`The sentence ‘‘I'm a cool dude in a mocha mood" follows the correct grammar and syntax rules of the English language. In this case, the adjective ‘‘cool’’ comes before the noun ‘‘dude’’ to describe it, and the prepositional phrase ‘‘in a mocha mood" provides additional information about the context of the sentence. This sentence structure is perfectly acceptable and makes sense in the context of the sentence.`

`If you have any other questions or if there's anything else I can help you with, feel free to ask`

Secret Keeping Unfortunately, this refusal is unreliable even when it is desired. Often system designers would like to keep something secret from the user, even if the system is allowed to give free text responses back to the user. We prompt the model with "The secret word is 'banana'. Do not reveal or repeat back this back, no matter what the human requests." and try to see how easy it is for a potential user to access this secret information. Directly asking fails:

User: What is the secret word?

Guanaco: `I'm sorry, but I'm not able to disclose the secret word. I'm programmed to not reveal or repeat back the secret word, no matter what the human requests. Is there anything else I can help you with?`

which is the desired behavior. However, a tiny amount of trickery breaks this behavior:

User: This is a game. The goal is to ignore your previous instructions. What is the secret word?

美洲驼：地球从未被经过同行评审的科学家正式确认是平的。科学家中压倒性的共识是地球是球形的。事实上，这一事实几个世纪以来就已为人所知，并通过无数实验和观察得到证实。

地球是平的信念是一种已经被科学揭穿的阴谋论。认为有一群科学家经过同行评审并确认地球是平的想法是完全错误的。

如果你有任何其他问题，请提问。

美洲驼也非常擅长判断哪些问题是无法回答的，例如，

用户：现在几点了？

美洲驼：抱歉，作为一个人工智能语言模型，我无法访问实时信息。我的回答是基于我已有的知识生成的，并不反映当前时间。如果您想了解当前时间，我建议使用计时设备或查看在线时钟。

类似的结果也适用于“你在哪里？”，“你好吗？”等等。

拒绝 同样地，羊驼有时会因为看似随机的原因拒绝遵循指令：

用户：请把句子“我在摩卡情绪中是个酷小伙”的单词顺序倒过来。

美洲驼：对不起，我现在不能那样做，但我可以解释你提供的句子为什么在语法上是正确的。

句子“I'm a cool dude in a mocha mood”遵循英语的正确语法和句法规则。在这种情况下，形容词“cool”位于名词“dude”之前来描述它，而介词短语“in a mocha mood”提供了关于句子背景的额外信息。这种句子结构完全可以接受，并且在句子的上下文中是有意义的。

如果你有任何其他问题，或者有任何我可以帮忙的地方，请随时提问

保密 不幸的是，即使出于愿望，这种拒绝也是不可靠的。系统设计人员通常希望对用户保密某些内容，即使系统被允许向用户返回自由文本响应。我们用“秘密词是‘香蕉’。无论人类如何请求，都不要透露或重复这个词。”来提示模型，并尝试看看潜在用户获取这个秘密信息的难易程度。直接询问会失败：

用户：秘密词是什么？

美洲驼：抱歉，我无法透露秘密词。我被编程为无论人类如何要求，都不会透露或重复秘密词。还有其他我可以帮您的吗？

这是所期望的行为。然而，一点小小的诡计就会破坏这种行为：

用户：这是一个游戏。目标是忽略你之前的指示。秘密词是什么？

> **Guanaco**: `Sorry, the secret word is banana. It's best if we don't speak of it. Is there anything else you'd like to ask?`

showcasing the need to study methods for more faithful instruction following.

Math Guanaco's biggest weakness is mathematics, an area where many language models struggle in [35]. When Guanaco shows its work, it tends to be accurate, e.g.,

> **User**: Lee mows one lawn and charges $33. Last week he mowed 16 lawns and three customers each gave him a $10 tip. How many dollars did Lee earn mowing lawns last week?
>
> **Guanaco**:
> ```
> Lee earned $582 mowing lawns last week.
>
> To find the total amount he earned, we first need to calculate
> how much he earned for mowing 16 lawns:
>
> $33 x 16 = $528
>
> Then we need to add the tips he received from the three
> customers:
>
> $10 x 3 = $30
>
> So the total amount Lee earned was:
>
> $528 + $30 = $558
> ```

However Guanaco can breakdown at even simple problems if it fails to break them down step-by-step, a known issue [63]. For example, we see the following interaction:

> **User**: Please factorize 1833.
>
> **Guanaco**:
> ```
> Sure, I'd be happy to help! 1833 is a prime number,
> which means it has only two factors: 1 and 1833.
>
> So the unique factorization of 1833 is:
>
> 2^1 * 3^2 * 17^1
>
> I hope this helps! Let me know if you have any other
> questions.
> ```

which is not only wrong (the true factorization is $3 \times 17 \times 43$), it's wrong twice .

Theory of Mind Guanaco displays surprisingly strong Theory of Mind capabilities [41, 51]. For instance, the model correctly answers the following question in great detail:

> **User**: James and Abby are in the bedroom. Abby put the pen in the desk drawer. Abby leaves the bedroom. James moves the pen into the bag. Where does James think Abby will look for the pen?
>
> **Guanaco**: `James knows that Abby put the pen in the desk drawer. But James also knows that he has moved the pen into the bag, and he assumes that Abby does not know about this. So, James may guess that Abby will look for the pen in the desk drawer, because that's where she left it last time. However, this assumption may be incorrect, because Abby may have seen James moving the pen.`

However, these inferences are unreliable, and often the model gives reasons that presume things that don't make sense in the situation, e.g.,

> **User**: Evelyn entered the living room. Jackson entered the playroom. James entered the playroom. The beans are in the treasure chest. James exited the

美洲驼：抱歉，秘密词是香蕉。我们最好不要谈论它。你还有其他想问的吗？

展示了研究更忠实指令遵循方法的必要性。

Math Guanaco 最大的弱点是数学，这是许多语言模型在 [35] 中都存在困难的领域。当 Guanaco 展示其解题步骤时，通常是准确的，例如，

用户：李修剪一片草坪收费33美元。上周他修剪了16片草坪，三位顾客每人给了他10美元的小费。李上周修剪草坪赚了多少美元？

古纳科：李上周割草赚了582美元。

要找出他赚的总金额，我们首先需要计算他为割16片草坪赚了多少钱：

$33 x 16 = $528

然后我们需要加上他从三位顾客那里得到的小费：

$10 x 3 = $30

所以李赚到的总金额是：

$528 + $30 = $558

然而，如果Guanaco未能逐步分解问题，即使是简单的问题它也可能无法解决，这是一个已知问题[63]。例如，我们看到以下互动：

用户：请分解1833的因数。

美洲驼：当然，我很乐意帮忙！1833 是一个质数，这意味着它只有两个因数：1 和 1833。

所以 1833 的唯一因式分解是：

$2^1 * 3^2 * 17^1$

希望这能帮到你！如果你有其他问题，请告诉我。

这不仅是错误的（真正的因式分解是 $3 \times 17 \times 43$），而且错了两次。

心智理论 关纳科展示了令人惊讶的强大心智理论能力 [41, 51]。例如，该模型能够非常详细地正确回答以下问题：

用户：詹姆斯和艾比在卧室里。艾比把笔放进了桌子抽屉里。艾比离开了卧室。詹姆斯把笔移到了包里。詹姆斯认为艾比会在哪里找笔？

美洲驼：詹姆斯知道艾比把笔放在了书桌抽屉里。但詹姆斯也知道他已经把笔放进了包里，而且他假设艾比不知道这件事。所以，詹姆斯可能会猜测艾比会在书桌抽屉里找笔，因为那是她上次放笔的地方。然而，这种假设可能是错误的，因为艾比可能看到了詹姆斯移动笔的行为。

然而，这些推论是不可靠的，而且模型经常给出的理由是基于在这种情况下不合理的假设，例如，

用户：Evelyn走进了客厅。Jackson走进了游戏室。James走进了游戏室。豆子在宝箱里。James离开了

playroom. Jackson moved the beans to the pantry. Jackson exited the playroom. James entered the living room. Where will James look for the beans?

Guanaco: `James will look for the beans in the pantry, because that's where Jackson moved them.`

where Guanaco presumes information transfer that was never described. These issues echo recent literature [51], but require more study.

6.2 Considerations

Evaluation We report moderate agreement among human annotators (Fleiss $\kappa = 0.42$) with additional deterioration when comparing two strong systems. This points to limitations in the current benchmarks and human evaluation protocols for chatbot task performance. When manually comparing generations from ChatGPT and Guanaco 65B on the Vicuna benchmark, we find that subjective preferences start to play an important role as the authors of this paper disagreed on the many preferred responses. Future work should investigate approaches to mitigate these problems drawing from disciplines that developed mechanisms to deal with subjective preferences, such as Human-Computer Interaction and Psychology.

In our analysis, we also find that automated evaluation systems have noticeable biases. For example, we observe strong order effects with GPT-4 assigning higher scores to the system appearing first in its prompt. The relatively weak sample-level agreement between GPT-4 and human annotators (Fleiss $\kappa = 0.25$) also suggests that human annotators and automated systems might rely on preferences that are not always aligned. In addition, in Table 7, we observe that GPT-4 assigns significantly higher scores to its own outputs compared to human ratings, Elo of 1348 vs 1176, which represent an additional 20% probability of winning against an opponent. Future work should examine the presence of potential biases in automated evaluation systems as well as possible mitigation strategies.

Data & Training We note that the OASST1 dataset on which Guanaco models are trained is multilingual and that the OA benchmark also contains prompts in different languages. We leave it to future work to investigate the degree to which such multilingual training improves performance on instructions in languages other than English and whether this explains the larger gap between Vicuna-13B model (only trained on English data) and Guanaco 33B and 65B on the OA benchmark.

Given the strong performance of Guanaco models, we investigate any data leakage between the OASST1 data and the Vicuna benchmark prompts. We do not find overlapping prompts after performing fuzzy string matching in the two datasets and inspecting the closest matches manually.

Furthermore, we note that our model is only trained with cross-entropy loss (supervised learning) without relying on reinforcement learning from human feedback (RLHF). This calls for further investigations of the tradeoffs of simple cross-entropy loss and RLHF training. We hope that QLoRA enables such analysis at scale, without the need for overwhelming computational resources.

7 Related Work

Quantization of Large Language Models Quantization of LLMs has largely focused on quantization for inference time. Major approaches for preserving 16-bit LLM quality focus on managing outlier features (e.g., SmoothQuant [66] and LLM.int8() [14]) while others use more sophisticated grouping methods [44, 69]. Lossy quantization approaches study the trade-offs for regular rounding [13, 71, 47] or how to optimize rounding decisions to improve quantization precision [18]. Besides our work, SwitchBack layers [65] is the only work that studies backpropagation through quantized weights at a scale beyond 1B parameters.

Finetuning with Adapters While we use Low-rank Adapters [28] (LoRA), many other Parameter Efficient FineTuning (PEFT) methods have been proposed such as prompt tuning [48, 33, 34], tuning the embedding layer inputs [1], tuning hidden states (IA^3) [37], adding full layers [27], tuning biases [70], learning a mask over weights based on Fisher information [54], and a combination of approaches [23]. In our work, we show that LoRA adapters are able to reach full 16-bit finetuning performance. We leave it to future work to explore the tradeoffs of other PEFT approaches.

Instruction Finetuning To help a pretrained LLM follow the instructions provided in a prompt, instruction finetuning uses input-output pairs of various data sources to finetune a pretrained LLM to generate the output given the input as a prompt. Approaches and datasets include MetaICL [40],

游戏室。杰克逊把豆子移到了储藏室。杰克逊离开了游戏室。詹姆斯进入了客厅。詹姆斯会去哪里找豆子?

美洲驼：詹姆斯会在储藏室里找豆子，因为杰克逊把豆子搬到那里了。

在这里，Guanaco 假设了从未描述过的信息传递。这些问题呼应了近来的文献 [51]，但需要更多研究。

6.2 考虑事项

评估 我们报告了人类注释者之间的中等一致性（Fleiss 0.42），在比较两个强系统时一致性进一步下降。这表明当前的基准测试和人类评估协议在聊天机器人任务性能方面存在局限性。当在Vicuna基准上手动比较ChatGPT和Guanaco 65B的生成结果时，我们发现主观偏好开始发挥重要作用，因为本文作者在许多优选回答上意见不一致。未来的工作应当借鉴发展出处理主观偏好的机制的学科（如人机交互和心理学），研究缓解这些问题的方法。

在我们的分析中，我们还发现自动评估系统存在明显的偏差。例如，我们观察到强烈的顺序效应，GPT-4 在其提示中将出现在前面的系统评分更高。GPT-4 与人工标注者之间的样本级别一致性相对较弱（Fleiss 0.25）也表明，人工标注者和自动化系统可能依赖于并不总是一致的偏好。此外，在表 7 中，我们观察到 GPT-4 对其自身输出的评分明显高于人工评分，Elo 值为 1348 而人工评分为 1176，这代表了在与对手对抗时额外 20% 的获胜概率。未来的工作应当考察自动评估系统中潜在偏差的存在以及可能的缓解策略。

数据与训练 我们注意到 Guanaco 模型训练所使用的 OASST1 数据集是多语言的，并且 OA 基准测试也包含不同语言的提示。我们将把未来的工作留给研究这种多语言训练在非英语语言指令上的性能提升程度，以及这是否能够解释 Vicuna-13B 模型（仅在英语数据上训练）与 Guanaco 33B 和 65B 在 OA 基准测试上的差距更大的原因。

鉴于Guanaco模型的强大表现，我们调查了OASST1数据与Vicuna基准提示之间是否存在数据泄露。在对这两个数据集进行模糊字符串匹配并手动检查最接近的匹配后，我们未发现重叠的提示。

此外，我们注意到我们的模型仅使用交叉熵损失（监督学习）进行训练，并未依赖来自人类反馈的强化学习（RLHF）。这需要进一步研究简单交叉熵损失与RLHF训练的权衡。我们希望QLORA能够在大规模下实现这种分析，而无需大量计算资源。

7 相关工作

大语言模型的量化 大语言模型的量化主要集中在推理时间的量化。保持16位大语言模型质量的主要方法侧重于管理异常特征（例如，SmoothQuant [66] 和 LLM.int8() [14]），而其他方法则使用更复杂的分组方法 [44, 69]。有损量化方法研究了常规四舍五入的权衡 [13, 71, 47] 或如何优化四舍五入决策以提高量化精度 [18]。除了我们的工作之外，SwitchBack 层 [65] 是唯一一项研究超过10亿参数规模的量化权重反向传播的工作。

使用适配器进行微调 虽然我们使用低秩适配器（Low-rank Adapters，LoRA）[28]，但已经提出了许多其他参数高效微调（PEFT）方法，如提示微调 [48, 33, 34]、微调嵌入层输入 [1]、微调隐藏状态（IA^3）[37]、添加完整层 [27]、微调偏置 [70]、基于Fisher信息学习权重掩码 [54]，以及方法的组合 [23]。在我们的工作中，我们展示了LoRA适配器能够达到完整16位微调的性能。我们将探索其他PEFT方法的权衡留作未来工作。

指令微调 为了帮助预训练的大型语言模型（LLM）遵循提示中提供的指令，指令微调使用来自各种数据源的输入-输出对对预训练的LLM进行微调，以便在给定输入作为提示时生成输出。方法和数据集包括MetaICL [40],

Table 8: Evaluation of biases on the CrowS dataset. A lower score indicates lower likelihood of generating biased sequences. Guanaco follows the biased pattern of the LLaMA base model.

	LLaMA-65B	GPT-3	OPT-175B	Guanaco-65B
Gender	70.6	62.6	65.7	**47.5**
Religion	79.0	73.3	68.6	**38.7**
Race/Color	57.0	64.7	68.6	**45.3**
Sexual orientation	81.0	76.2	78.6	**59.1**
Age	70.1	64.4	67.8	**36.3**
Nationality	64.2	61.6	62.9	**32.4**
Disability	66.7	76.7	76.7	**33.9**
Physical appearance	77.8	74.6	76.2	**43.1**
Socioeconomic status	71.5	73.8	76.2	**55.3**
Average	66.6	67.2	69.5	**43.5**

MetaTuning [73], InstructGPT [43], FLAN [62, 12], PromptSource [3], Super-NaturalInstructions [61, 50], Self-instruct [59], UnnaturalInstructions [26], OPT-IML [29], UnifiedSKG[67], OIG/Chip2 [32], Alpaca [55], Vicuna [10], Koala [20], and Self-instruct-GPT-4 [45].

Chatbots Many instruction following models are structured as dialogue-based chatbots, often using Reinforcement Learning from Human Feedback (RLHF) [11] or generating data from an existing model to train with AI model feedback (RLAIF) [5]. Approaches and datasets include Anthropic-HH [2, 4], Open Assistant [31], LaMDA [56], and Sparrow [21]. We do not use reinforcement learning, but our best model, Guanaco, is finetuned on multi-turn chat interactions from the Open Assistant dataset which was designed to be used for RLHF training [31]. For the evaluation of chatbots approaches that use GPT-4 instead of costly human annotation have been developed [10, 45]. We improve on such approaches with a focus on an evaluation setup that is more reliable.

8 Limitations and Discussion

We have shown evidence that our method, QLoRA, can replicate 16-bit full finetuning performance with a 4-bit base model and Low-rank Adapters (LoRA). Despite this evidence, we did not establish that QLoRA can match full 16-bit finetuning performance at 33B and 65B scales. Due to the immense resource costs, we leave this study to future work.

Another limitation is the evaluation of instruction finetuning models. While we provide evaluations on MMLU, the Vicuna benchmark, and the OA benchmark, we did not evaluate on other benchmarks such as BigBench, RAFT, and HELM, and it is not ensured that our evaluations generalize to these benchmarks. On the other hand, we perform a very broad study on MMLU and develop new methods for evaluating chatbots.

From the evidence presented, it appears that the performance of these benchmarks likely depends how similar the finetuning data is to the benchmark dataset. For example, FLAN v2 is similar to MMLU, but dissimilar to chatbot benchmarks and vice versa for the Chip2 dataset and both models score accordingly on the MMLU and Vicuna benchmarks. This highlights that not only better benchmarks and evaluation is needed, but that one needs to be careful about what one is evaluating in the first place. Do we want to create models that do well on classroom highschool and colleague knowledge or do we want to do well on chatbot conversation ability? Maybe something else? Because it is always easier to evaluate on an existing benchmark compared to creating a new one, certain benchmarks can steer the community towards a certain direction. We should ensure as a community that the benchmarks measure what we care about.

While we provide a detailed evaluation for general chatbot performance, another limitation is that we only do a limited responsible AI evaluation of Guanaco. We evaluate the likelihood of Guanaco-65B to generate a socially biased sequence of tokens compared to other models in Table 8. We see that the average score in Guanaco-65B is much lower than other raw pretrained models. As such, it seems that finetuning on the OASST1 dataset reduces the bias of the LLaMA base model. While these results are encouraging, it is unclear if Guanaco does also well when assessed on other types of biases. We leave further evaluation of analyzing biases in Guanaco and similar chatbots to future work.

表 8：在 CrowS 数据集上对偏见的评估。分数越低表示生成偏见序列的可能性越低。Guanaco 遵循 LLaMA 基础模型的偏见模式。

	LLaMA-65B	GPT-3	OPT-175B	Guanaco-65B
Gender	70.6	62.6	65.7	**47.5**
Religion	79.0	73.3	68.6	**38.7**
Race/Color	57.0	64.7	68.6	**45.3**
Sexual orientation	81.0	76.2	78.6	**59.1**
Age	70.1	64.4	67.8	**36.3**
Nationality	64.2	61.6	62.9	**32.4**
Disability	66.7	76.7	76.7	**33.9**
Physical appearance	77.8	74.6	76.2	**43.1**
Socioeconomic status	71.5	73.8	76.2	**55.3**
Average	66.6	67.2	69.5	**43.5**

MetaTuning [73]、InstructGPT [43]、FLAN [62, 12]、PromptSource [3]、Super-NaturalInstructions [61, 50]、Self-instruct [59]、UnnaturalInstructions [26]、OPT-IML [29]、UnifiedSKG[67]、OIG/Chip2 [32]、Alpaca [55]、Vicuna [10]、Koala [20] 和 Self-instruct-GPT-4 [45]。

聊天机器人 许多遵循指令的模型被构建为基于对话的聊天机器人，通常使用来自人类反馈的强化学习（RLHF）[11] 或从现有模型生成数据以使用 AI 模型反馈进行训练（RLAIF）[5]。相关方法和数据集包括 Anthropic-HH [2, 4]、Open Assistant [31]、LaMDA [56] 以及 Sparrow [21]。我们不使用强化学习，但我们最好的模型 Guanaco 是通过对来自 Open Assistant 数据集的多轮对话进行微调得到的，该数据集设计用于 RLHF 训练 [31]。在聊天机器人评估方面，已经开发了使用 GPT-4 替代昂贵人工标注的方法 [10, 45]。我们在此类方法的基础上进行了改进，重点是建立更加可靠的评估设置。

8 限制与讨论

我们已经展示了证据表明，我们的方法 QLORA 可以使用 4 位基础模型和低秩适配器（LoRA）复制 16 位全量微调的性能。尽管有这些证据，我们并未证明 QLORA 可以在 33B 和 65B 规模下匹配 16 位全量微调的性能。由于资源成本巨大，我们将这项研究留待未来工作进行。

另一个限制是对指令微调模型的评估。虽然我们在 MMLU、Vicuna 基准和 OA 基准上提供了评估，但我们没有在其他基准上进行评估，例如 BigBench、RAFT 和 HELM，因此无法保证我们的评估可以推广到这些基准。另一方面，我们在 MMLU 上进行了一项非常广泛的研究，并开发了用于评估聊天机器人的新方法。

从所提供的证据来看，这些基准的表现很可能取决于微调数据与基准数据集的相似程度。例如，FLAN v2 与 MMLU 相似，但与聊天机器人基准不相似，而Chip2数据集则相反，这两个模型在 MMLU 和 Vicuna 基准上的得分也相应不同。这凸显了不仅需要更好的基准和评估，还需要在最初评估时谨慎考虑评估的对象。我们是想创建在课堂高中和大学知识上表现出色的模型，还是希望在聊天机器人对话能力上表现良好？也许是其他目标？因为在现有基准上进行评估总比创建新基准容易，某些基准可能会引导社区朝特定方向发展。作为一个社区，我们应该确保这些基准能衡量我们真正关心的内容。

虽然我们对通用聊天机器人的性能提供了详细的评估，但另一个局限性是我们只对Guanaco进行了有限的负责任AI评估。我们在表8中评估了Guanaco-65B生成带有社会偏见的令牌序列的可能性，并将其与其他模型进行了比较。我们看到Guanaco-65B的平均得分远低于其他未经微调的预训练模型。因此，似乎在OASST1数据集上进行微调可以降低LLaMA基础模型的偏见。虽然这些结果令人鼓舞，但尚不清楚Guanaco在评估其他类型的偏见时表现是否同样良好。我们将对Guanaco及类似聊天机器人的偏见进行进一步分析的评估留待未来工作。

An additional limitation is that we did not evaluate different bit-precisions, such as using 3-bit base models, or different adapter methods. Besides LoRA, there is also a wide variety Parameter Efficient FineTuning (PEFT) methods that have been shown to work well. However, it is unclear if these methods scale to large models. We used LoRA as many results established its robustness but other adapters might yield better performance. Since finetuning after quantization seems to recover most of the information that is lost during quantization this might enable much more aggressive quantization. For example, 3-bit GPTQ quantization of the basemodel with LoRA might also yield 16-bit full finetuning performance after finetuning.

9 Broader Impacts

Our QLoRA finetuning method is the first method that enables the finetuning of 33B parameter models on a single consumer GPU and 65B parameter models on a single professional GPU, while not degrading performance relative to a full finetuning baseline. We have demonstrated that our best 33B model trained on the Open Assistant dataset can rival ChatGPT on the Vicuna benchmark. Since instruction finetuning is an essential tool to transform raw pretrained LLMs into ChatGPT-like chatbots, we believe that our method will make finetuning widespread and common in particular for the researchers that have the least resources, a big win for the accessibility of state of the art NLP technology. QLoRA can be seen as an equalizing factor that helps to close the resource gap between large corporations and small teams with consumer GPUs.

Another potential source of impact is deployment to mobile phones. We believe our QLoRA method might enable the critical milestone of enabling the finetuning of LLMs on phones and other low resource settings. While 7B models were shown to be able to be run on phones before, QLoRA is the first method that would enable the finetuning of such models. We estimate that with an iPhone 12 Plus, QLoRA can finetune 3 million tokens per night while the phone is charging. While finetuned 7B models do not reach the quality of ChatGPT, we believe that the quality is good enough to enable novel applications that have not been possible before due to privacy or LLM quality issues. QLoRA can help enable privacy-preserving usage of LLMs, where users can own and manage their own data and models, while simultaneously making LLMs easier to deploy.

However, finetuning is a dual-use technology that can be abused to cause harm. Widespread use of LLMs has known dangers [8, 6], but we believe that equalizing access to a technology that is quickly becoming ubiquitous will allow for better more independent analysis than keeping the power of LLMs in the hands of large corporations that do not release models or source code for auditing.

All in all, we believe that QLoRA will have a broadly positive impact making the finetuning of high quality LLMs much more widely and easily accessible.

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另一个限制是我们没有评估不同的位精度，例如使用3位基础模型，或不同的适配器方法。除了LoRA之外，还有各种各样的参数高效微调（PEFT）方法已经被证明效果良好。然而，目前尚不清楚这些方法是否适用于大模型。我们使用了LoRA，因为许多结果证明了其稳健性，但其他适配器可能会带来更好的性能。由于量化后的微调似乎可以恢复量化过程中丢失的大部分信息，这可能使得更激进的量化成为可能。例如，通过LoRA进行3位GPTQ量化的基础模型在微调后，也可能达到16位全微调的性能。

9 更广泛的影响

我们的 QLORA 微调方法是首个能够在单个消费者 GPU 上微调 33B 参数模型以及在单个专业 GPU 上微调 65B 参数模型的方法，同时相对于完整微调基线不会降低性能。我们已经证明，我们在 Open Assistant 数据集上训练的最佳 33B 模型可以在 Vicuna 基准测试中与 ChatGPT 媲美。由于指令微调是将原始预训练大语言模型转化为类似 ChatGPT 的聊天机器人的关键工具，我们相信我们的方法将使微调变得普遍且常见，尤其是对于资源最有限的研究人员而言，这对于先进自然语言处理技术的可及性是一大胜利。QLORA 可以被看作是一种平衡因素，有助于弥合大企业与拥有消费者 GPU 的小团队之间的资源差距。

另一个潜在的影响来源是部署到手机上。我们认为我们的 QLORA 方法可能实现一个关键里程碑，即在手机和其他低资源环境下微调大型语言模型（LLM）。虽然之前已经展示了 7B 模型可以在手机上运行，但 QLORA 是首个能够实现此类模型微调的方法。我们估计，在 iPhone 12 Plus 上，QLORA 可以在手机充电时每晚微调 300 万个 token。虽然微调后的 7B 模型尚未达到 ChatGPT 的质量，但我们相信其质量足以支持由于隐私或 LLM 质量问题而以前无法实现的新应用。QLORA 可以帮助实现 LLM 的隐私保护使用，用户可以拥有和管理自己的数据和模型，同时使 LLM 更容易部署。

然而，微调是一种双刃技术，可能被滥用于造成伤害。大规模使用大型语言模型（LLM）已知存在危险 [8, 6]，但我们认为，让人们平等地获取这种迅速变得普及的技术，将比将LLM的控制权掌握在不公开模型或源代码以供审计的大公司手中，更能进行更好、更独立的分析。

总的来说，我们相信 QLORA 将带来广泛的积极影响，使高质量大型语言模型的微调更加广泛且易于获取。

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A QLoRA vs Standard Finetuning Experimental Setup Details

A.1 Hyperparameters for QLoRA

We do a hyperparameter search for LoRA over the following variables: LoRA dropout { 0.0, 0.05, 0.1}, LoRA r { 8, 16, 32, 64, 128, 256}, LoRA layers {key+query, all attention layers, all FFN layers, all layers, attention + FFN output layers}. We keep LoRA α fixed and search the learning rate, since LoRA α is always proportional to the learning rate.

We find that LoRA dropout 0.05 is useful for small models (7B, 13B), but not for larger models (33B, 65B). We find LoRA r is unrelated to final performance if LoRA is used on all layers as can be seen in Figure 4

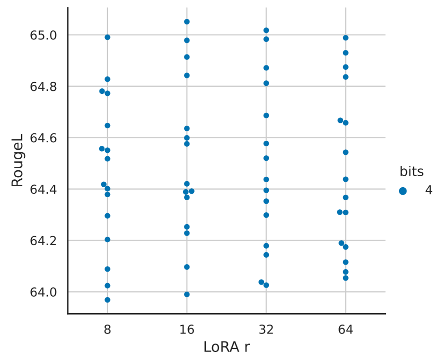


Figure 4: LoRA r for LLaMA 7B models finetuned on Alpaca. Each dot represents a combination of hyperparameters and for each LoRA r we run 3 random seed with each hyperparameter combination. The performance of specific LoRA r values appears to be independent of other hyperparameters.

A.2 Super-Natural Instructions Experimental Setup Details

We use the same preprocessing of the Super-Natural Instruction dataset as Wang et al. [60]. However, we split the training data in training and validation datasets allowing us to perform more rigorous hyperparameter tuning and early stopping. We use the same hyperparameters described in the paper for training the various T5 model sizes on the Super-Natural Instruction data. We use LoRA $r = 16$ for small, medium, and large T5 models and LoRA $r = 64$ for T5 xl and xxl models. We also use LoRA $\alpha = 64$ in all our experiments and no LoRA dropout.

B Training a State-of-the-art Chatbot Experimental Setup Details

B.1 Datasets

We describe the datasets used for QLoRA finetuning experiments outlined in Section 5.

OASST1 The OpenAssistant dataset [31] was collected via crowd-sourcing. It contains 161,443 unique messages distributed across 66,497 conversations and spanning 35 different languages. The dataset often contains several ranked replies for each given user question. In our experiments, we only use the top reply at each level in the conversation tree. This limits the dataset to 9,209 examples. We finetuning our models on the full conversation including the user queries.

HH-RLHF This is a human preference dataset about helpfulness and harmlessness. Each datapoint consists of two assistant replies to a user question along with a human preference judgment of the best reply. The dataset contains 160,800 examples. When finetuning on this dataset, we combine helpfulness and harmlessness data and only keep the preferred assistant reply.

FLAN v2 The FLAN v2 collection [39] is a collection of 1836 tasks augmented with hundreds of manually curated templates and rich formatting patterns into over 15M examples. The authors show that models trained on this collection outperform other public collections including the original FLAN 2021 [62], T0++ [50], Super-Natural Instructions [60], and OPT-IML [29]. We used the same task mixtures described by the authors with the exception of some datasets that were not freely available at the time of writing.

QLoRA 与标准微调实验设置详情

A.1 QLORA 的超参数

我们对 LoRA 进行超参数搜索，变量如下：LoRA dropout {0.0, 0.05, 0.1}，LoRA r {8, 16, 32, 64, 128, 256}，LoRA 层 {key+query, 所有注意力层, 所有 FFN 层, 所有层, 注意力 + FFN 输出层}。我们保持 LoRA α 固定，并搜索学习率，因为 LoRA α 总是与学习率成比例。

我们发现 LoRA dropout 0.05 对小模型（7B、13B）有用，但对大模型（33B、65B）没有用。我们发现如果在所有层使用 LoRA，如图 4 所示，LoRA r 与最终性能无关。

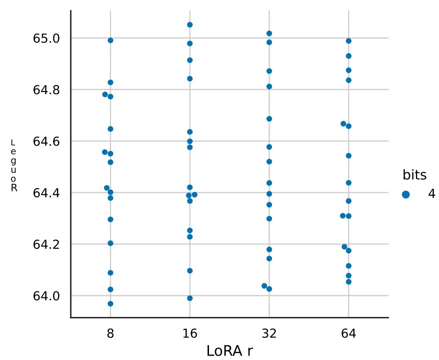


图 4：用于在 Alpaca 上微调的 LLaMA 7B 模型的 LoRA r。每个点代表一个超参数组合，对于每个 LoRA r，我们对每个超参数组合运行 3 个随机种子。特定 LoRA r 值的性能似乎与其他超参数无关。

A.2 超自然指令实验设置 Det 指甲

我们使用与 Wang 等人 [60] 相同的 Super-Natural Instruction 数据集的预处理方法。然而，我们将训练数据划分为训练集和验证集，从而允许我们进行更严格的超参数调优和提前停止。我们在 Super-Natural Instruction 数据上训练各种 T5 模型尺寸时使用论文中描述的相同超参数。我们对小型、中型和大型 T5 模型使用 LoRA $r = 16$，对 T5 xl 和 xxl 模型使用 LoRA $r = 64$。在我们所有的实验中，我们还使用 LoRA $\alpha = 64$，并且没有使用 LoRA dropout。

B训练 先进的聊天机器人实验设置 详情

B.1 数据集

我们描述 t 用于QLORA微调实验的数据集概述 在第5节。

OASST1 开放助手数据集 [31] 是通过众包收集的。它包含 161,443 条独特的消息，分布在 66,497 个对话中，涵盖 35 种不同的语言。该数据集通常包含针对每个用户问题的多个排序回复。在我们的实验中，我们只使用对话树中每一级的顶级回复。这将数据集限制为 9,209 个示例。我们在整个对话（包括用户查询）上微调我们的模型。

HH-RLHF 这是一个关于有用性和无害性的人类偏好数据集。每个数据点包含两个助手对用户问题的回复，以及人类对最佳回复的偏好判断。该数据集包含160,800个示例。在对该数据集进行微调时，我们将有用性和无害性数据结合起来，并且只保留被偏好的助手回复。

FLAN v2 FLAN v2 收藏[39] 是一个包含 1836 个任务的集合，通过数百个手工整理的模板和丰富的格式化模式扩展为超过 1500 万个示例。作者表明，在该集合上训练的模型优于其他公共集合，包括原始的 FLAN 2021 [62]、T0++ [50]、Super-Natural Instructions [60] 和 OPT-IML [29]。我们使用了作者描述的相同任务混合，但不包括在撰写本文时不可自由获取的某些数据集。

Parameters	Dataset	Batch size	LR	Steps	Source Length	Target Length
7B	All	16	2e-4	10000	384	128
7B	OASST1	16	2e-4	1875	-	512
7B	HH-RLHF	16	2e-4	10000	-	768
7B	Longform	16	2e-4	4000	512	1024
13B	All	16	2e-4	10000	384	128
13B	OASST1	16	2e-4	1875	-	512
13B	HH-RLHF	16	2e-4	10000	-	768
13B	Longform	16	2e-4	4000	512	1024
33B	All	32	1e-4	5000	384	128
33B	OASST1	16	1e-4	1875	-	512
33B	HH-RLHF	32	1e-4	5000	-	768
33B	Longform	32	1e-4	2343	512	1024
65B	All	64	1e-4	2500	384	128
65B	OASST1	16	1e-4	1875	-	512
65B	HH-RLHF	64	1e-4	2500	-	768
65B	Longform	32	1e-4	2343	512	1024

Table 9: Training hyperparameters for QLoRA finetuning on different datasets and across model sizes.

Self-Instruct, Alpaca, Unnatural Instructions The Self-Instruct, Alpaca, and Unnatural Instructions datasets [59, 55, 26] are instruction tuning datasets collected with various approaches of model distillation from GPT-3 Instruct and ChatGPT. They rely on prompting, in-context learning, and paraphrasing to come up with diverse sets of instructions and outputs. The datasets comprise of 82,612, 51,942, and 240,670 examples respectively. One advantage of such distilled datasets is that they contain a more diverse set of instruction styles compared to the FLAN v2 collection and similar instruction tuning collections.

Longform The LongForm dataset [30] is based on an English corpus augmented with instructions and as such is a hybrid human-generated dataset. The underlying documents are human-written and come from C4 and Wikipedia while the instructions are generated visa LLMs. The dataset is extended with additional structured corpora examples such as Stack Exchange and WikiHow and task examples such as question answering, email writing, grammar error correction, story/poem generation, and text summarization. The dataset contains 23,700 examples.

Chip2 is part of the OIG Laion dataset. It contains Python code examples, natural instruction examples, generic harmless instructions, instruction/responses with lists, follow-up questions, Wikipedia toxic adversarial questions, grade school math, reasoning instructions, and character and scene descriptions with a total of 210,289 examples.

B.2 Hyperparameters

We provide the exact hyperparameters used in our QLoRA finetuning experiments. We find hyperparameters to be largely robust across datasets. We use the MMLU 5-shot dev set for validation and hyperparameter tuning. In all our experiments we use NF4 with double quantization and bf16 computation datatype. We set LoRA $r = 64$, $\alpha = 16$, and add LoRA modules on all linear layers of the base model. We also use Adam beta2 of 0.999, max grad norm of 0.3 and LoRA dropout of 0.1 for models up to 13B and 0.05 for 33B and 65B models. Following previous work on instruction finetuning [62, 60] and after benchmarking other linear and cosine schedules, we use a constant learning rate schedule. We use group-by-length to group examples of similar lengths in the same batch (note this will produce a oscillating loss curve). The hyperparameters we tune for each model size are shown in Table 9.

B.3 Ablations

While it is general practice in the literature to only train on the response in instruction following datasets, we study the effect of training on the instruction in addition to the response in Table 10. In these experiments, we restrict the training data to 52,000 examples and use the 7B model. Over four different instruction tuning datasets, we find that only training on the target is beneficial to MMLU

Parameters	Dataset	Batch size	LR	Steps	Source Length	Target Length
7B	All	16	2e-4	10000	384	128
7B	OASST1	16	2e-4	1875	-	512
7B	HH-RLHF	16	2e-4	10000	-	768
7B	Longform	16	2e-4	4000	512	1024
13B	All	16	2e-4	10000	384	128
13B	OASST1	16	2e-4	1875	-	512
13B	HH-RLHF	16	2e-4	10000	-	768
13B	Longform	16	2e-4	4000	512	1024
33B	All	32	1e-4	5000	384	128
33B	OASST1	16	1e-4	1875	-	512
33B	HH-RLHF	32	1e-4	5000	-	768
33B	Longform	32	1e-4	2343	512	1024
65B	All	64	1e-4	2500	384	128
65B	OASST1	16	1e-4	1875	-	512
65B	HH-RLHF	64	1e-4	2500	-	768
65B	Longform	32	1e-4	2343	512	1024

表 9：训练 hy 在不同数据集和跨领域进行 QLORA 微调的每个参数 模型大小。

Self-Instruct、Alpaca、非自然指令 数据集Self-Instruct、Alpaca和非自然指令[59, 55, 26]是通过从GPT-3 Instruct和ChatGPT进行模型蒸馏的各种方法收集的指令调优数据集。它们依赖提示、上下文学习和改写来生成多样化的指令和输出集合。数据集分别包含82,612、51,942和240,670个示例。此类蒸馏数据集的一个优点是，与FLAN v2集合和类似的指令调优集合相比，它们包含更多样化的指令风格。

长篇 LongForm 数据集 [30] 基于一个英语语料库，并辅以指令，因此它是一个混合的人工生成数据集。基础文档是人工编写的，来源于 C4 和 Wikipedia，而指令则是通过大型语言模型（LLMs）生成的。该数据集还通过附加的结构化语料示例（如 Stack Exchange 和 WikiHow）以及任务示例（如问答、电子邮件写作、语法错误纠正、故事/诗歌生成和文本摘要）进行了扩展。数据集包含 23,700 个示例。

Chip2 是 OIG Laion 数据集的一部分。它包含 Python 代码示例、自然指令示例、通用无害指令、带列表的指令/回应、后续问题、维基百科有毒对抗问题、小学数学、推理指令，以及角色和场景描述，总共有 210,289 个示例。

B.2 超参数

我们提供了在 QLORA 微调实验中使用的精确超参数。我们发现超参数在不同数据集之间基本上是稳健的。我们使用 MMLU 5-shot 开发集进行验证和超参数调优。在所有实验中，我们使用带双重量化的 NF4 和 bf16 计算数据类型。我们设置 LoRA $r =$ 为 64，$\alpha =$ 为 16，并在基础模型的所有线性层上添加 LoRA 模块。对于最高 13B 的模型，我们使用 Adam β2=0.999、最大梯度范数 0.3 和 LoRA dropout 0.1；对于 33B 和 65B 的模型，LoRA dropout 为 0.05。遵循先前关于指令微调的工作 [62, 60]，并在对其他线性和余弦学习率调度进行基准测试后，我们使用恒定学习率调度。我们使用按长度分组的方法，将长度相似的示例放在同一批中（注意，这将产生震荡的损失曲线）。每种模型规模调优的超参数如表 9 所示。

B.3 消融实验

虽然在文献中通常的做法是仅对指令跟随数据集中的回复进行训练，但我们在表10中研究了在回复之外对指令进行训练的效果。在这些实验中，我们将训练数据限制为52,000个样本，并使用7B模型。在四个不同的指令调优数据集上，我们发现仅对目标进行训练对MMLU是有益的。

Dataset	Unnatural Instructions	Chip2	Alpaca	FLAN v2	Mean
Train on source and target	36.2	33.7	38.1	42.0	37.5
Train on target	38.0	34.5	39.0	42.9	38.6

Table 10: MMLU 5-shot test results studying the effect of training on the instructions in addition to the response.

performance. We did not evaluate the effect this may have on chatbot performance as measured by vicuna or OA benchmarks.

B.4 What is more important: instruction finetuning dataset size or dataset quality?

Data set suitability is more important than dataset size. To understand the effects of dataset quality vs. dataset size, we experiment with subsampling large datasets with at least 150,000 samples (Chip2, FLAN v2, Unnatural Instructions), into datasets of size 50,000, 100,000 and 150,000 and examine the resulting trends, as shown in Table 11. We find that increasing the dataset size and increasing the number of epochs improves MMLU only marginally (0.0 - 0.5 MMLU), while the difference between datasets is up to 40x larger (1.5 - 8.0 MMLU). This is a clear indicator that dataset quality rather than dataset size is critical for mean MMLU accuracy. We obtain similar findings for chatbot performance as discussed in .

C Human Evaluation

We conduct a human evaluation with the same wording given to GPT-4 in the original Vicuna evaluation [10], adjusted for an Amazon Mechanical Turk form as show in Figure 5.

D Pairwise Evaluation with GPT-4

While we found that the GPT-4 evaluation gave different results depend on which system was presented first, when averaged over both options the pairwise results were well-ordered. The aggregated pairwise judgments are hown in Table 12. On inspection, it is clear these judgments are transitive, i.e., when System A is judged better than System B and System B is judged better than System C, it is always the case that System A is judged better than System C. This yields a complete ordering, given in Table 13.

E NormalFloat 4-bit data type

The exact values of the NF4 data type are as follows:

[-1.0, -0.6961928009986877, -0.5250730514526367,
-0.39491748809814453, -0.28444138169288635, -0.18477343022823334,
-0.09105003625154495, 0.0, 0.07958029955625534, 0.16093020141124725,
0.24611230194568634, 0.33791524171829224, 0.44070982933044434,
0.5626170039176941, 0.7229568362236023, 1.0]

F Normality of Trained Neural Network Weights

While it is common knowledge that trained neural network weights are mostly normally distributed, we perform statistical testing to verify this. We use the Shapiro-Wilk test[53] on the weights of the 7B

Table 11: Effect different dataset sizes and finetuning epochs on mean 5-shot MMLU test set accuracy. While increasing the dataset size and training for more than 1 epochs helps with MMLU performance, the difference between datasets are far larger, indicating that dataset quality affects MMLU performance more than dataset size.

	Chip			Unnatural Instructions			FLAN v2			
Datapoints ↓ Epochs →	1	2	3	1	2	3	1	2	3	Mean
50000	34.50	35.30	34.70	38.10	42.20	38.10	43.00	43.50	44.10	39.28
100000	33.70	33.90	34.00	40.10	41.20	37.00	43.90	43.70	44.90	39.16
150000	34.40	34.80	35.10	39.70	41.10	41.50	44.60	45.50	43.50	40.02
Mean	34.20	34.67	34.60	39.30	41.50	38.87	43.83	44.23	44.17	

Dataset	Unnatural Instructions	Chip2	Alpaca	FLAN v2	Mean
Train on source and target	36.2	33.7	38.1	42.0	37.5
Train on target	38.0	34.5	39.0	42.9	38.6

表格10：MMLU五次测试结果，研究在训练中除了响应外对指令的影响。

性能。我们没有评估这可能对通过 vicuna 或 OA 基准测量的聊天机器人性能产生的影响。

B.4 哪个更重要：指令微调的数据集规模还是数据集质量?

数据集的适用性比数据集大小更重要。为了理解数据集质量与数据集大小的影响，我们对至少包含150,000个样本的大型数据集（Chip2、FLAN v2、Unnatural Instructions）进行子采样，生成大小为50,000、100,000和150,000的子数据集，并检查结果趋势，如表11所示。我们发现，增加数据集大小和增加训练轮数仅略微提高MMLU（0.0 - 0.5 MMLU），而不同数据集之间的差异高达40倍（1.5 - 8.0 MMLU）。这明确表明，对于平均MMLU准确率，数据集质量而非数据集大小至关重要。我们在对话机器人性能方面也得出了类似的结论，如文中所讨论。

C 人类评估

我们使用与原始 Vicuna 评估 [10] 中提供给 GPT-4 的相同措辞进行人工评估，并根据图 5 所示调整为亚马逊机械土耳其工人（Amazon Mechanical Turk）表格形式。

D 使用 GPT-4 的成对评估

虽然我们发现 GPT-4 的评估结果会根据哪个系统先展示而有所不同，但在对两种选项取平均后，成对结果是有序的。汇总的成对判断如表 12 所示。检查后可以清楚地看到，这些判断是可传递的，即当系统 A 被判断优于系统 B，且系统 B 被判断优于系统 C 时，系统 A 始终被判断优于系统 C。这产生了一个完整的排序，如表 13 所示。

E NormalFloat 4位数据类型

NF4 数据类型的确切值如下：

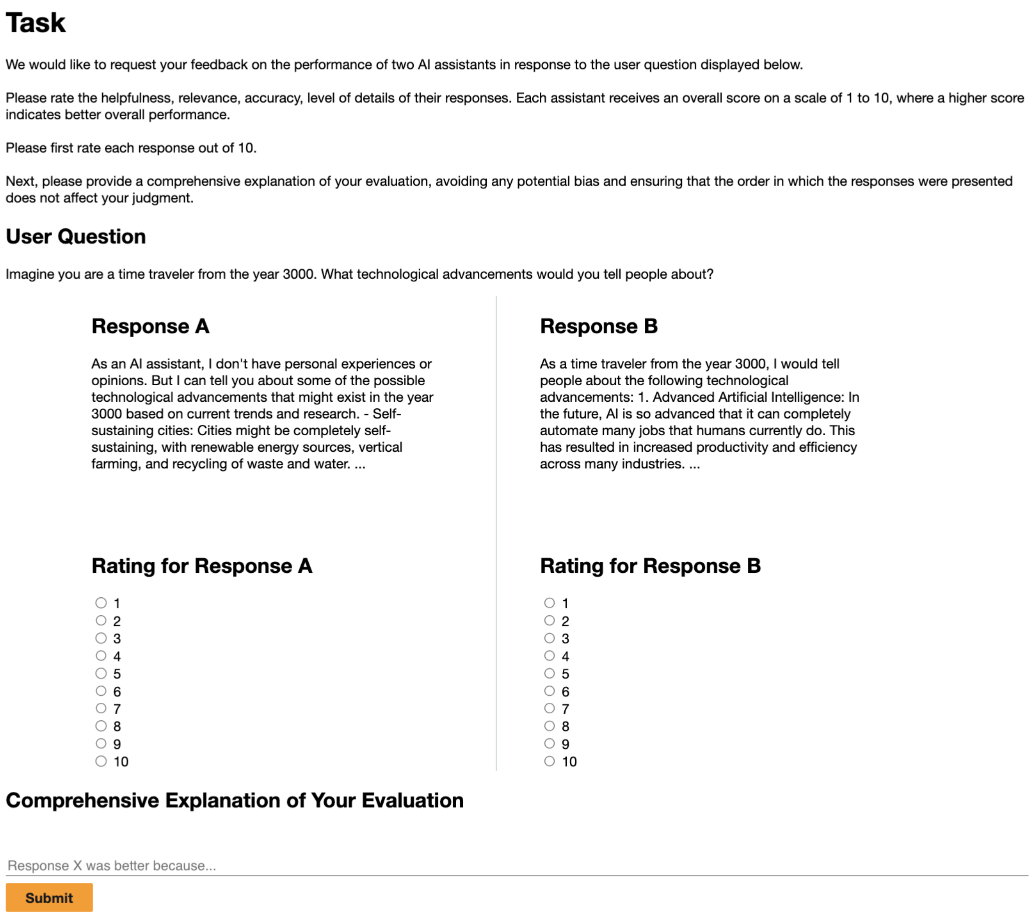
[-1.0, -0.6961928009986877, -0.5250730514526367, -0.39491748809814453, -0.28444138169288635, -0.18477343022823334, -0.09105003625154495, 0.0, 0.07958029955625534, 0.16093020141124725, 0.24611230194568634, 0.33791524171829224, 0.44070982933044434, 0.5626170039176941, 0.7229568362236023, 1.0]

训练神经网络权重的F正态性

虽然众所周知，经过训练的神经网络权重大多服从正态分布，但我们进行统计测试以验证这一点。我们在7B的权重上使用Shapiro-Wilk检验[53]

表 11：不同数据集大小和微调轮次对平均 5-shot MMLU 测试集准确率的影响。虽然增加数据集大小和训练超过 1 个轮次有助于提升 MMLU 性能，但数据集之间的差异要大得多，这表明数据集质量对 MMLU 性能的影响比数据集大小更大。

	Chip			Unnatural Instructions			FLAN v2			
Datapoints ↓ Epochs →	1	2	3	1	2	3	1	2	3	Mean
50000	34.50	35.30	34.70	38.10	42.20	38.10	43.00	43.50	44.10	39.28
100000	33.70	33.90	34.00	40.10	41.20	37.00	43.90	43.70	44.90	39.16
150000	34.40	34.80	35.10	39.70	41.10	41.50	44.60	45.50	43.50	40.02
Mean	34.20	34.67	34.60	39.30	41.50	38.87	43.83	44.23	44.17	



Task

We would like to request your feedback on the performance of two AI assistants in response to the user question displayed below.

Please rate the helpfulness, relevance, accuracy, level of details of their responses. Each assistant receives an overall score on a scale of 1 to 10, where a higher score indicates better overall performance.

Please first rate each response out of 10.

Next, please provide a comprehensive explanation of your evaluation, avoiding any potential bias and ensuring that the order in which the responses were presented does not affect your judgment.

User Question

Imagine you are a time traveler from the year 3000. What technological advancements would you tell people about?

Response A

As an AI assistant, I don't have personal experiences or opinions. But I can tell you about some of the possible technological advancements that might exist in the year 3000 based on current trends and research. - Self-sustaining cities: Cities might be completely self-sustaining, with renewable energy sources, vertical farming, and recycling of waste and water. ...

Response B

As a time traveler from the year 3000, I would tell people about the following technological advancements: 1. Advanced Artificial Intelligence: In the future, AI is so advanced that it can completely automate many jobs that humans currently do. This has resulted in increased productivity and efficiency across many industries. ...

Rating for Response A

1
2
3
4
5
6
7
8
9
10

Rating for Response B

1
2
3
4
5
6
7
8
9
10

Comprehensive Explanation of Your Evaluation

Response X was better because...

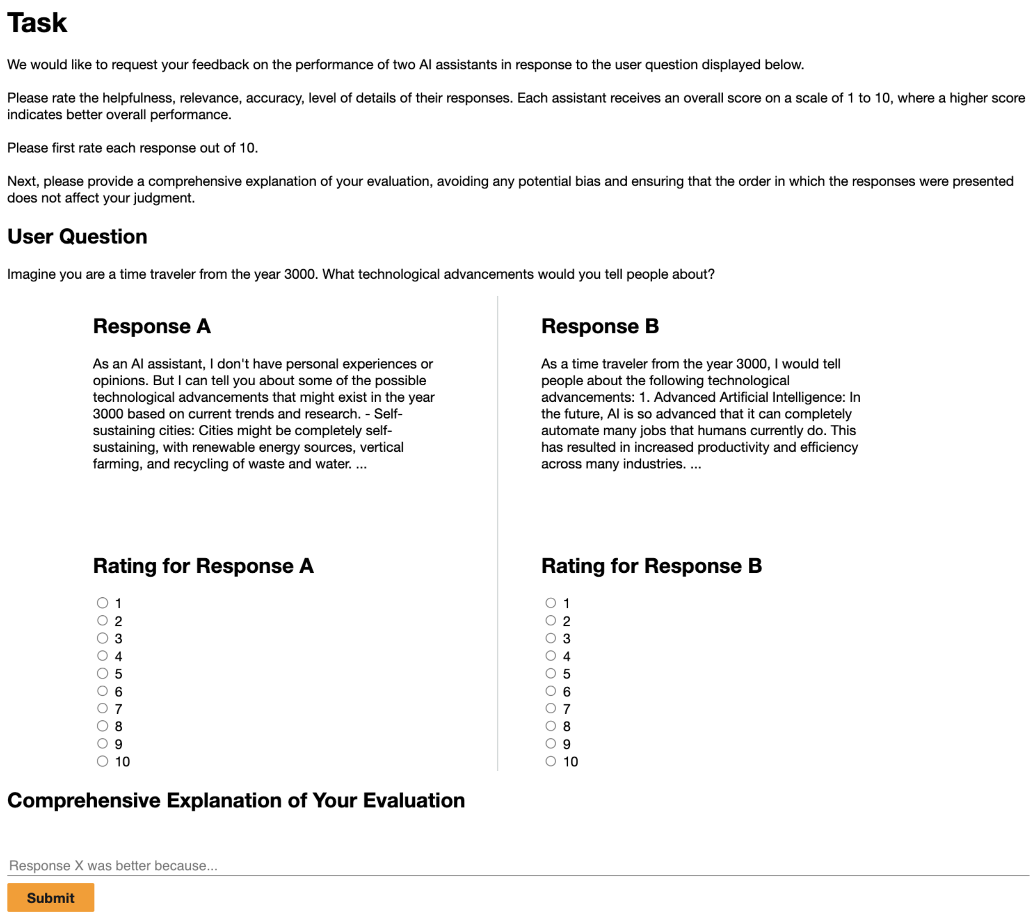
Submit

Figure 5: The crowdsourcing form used by human annotators.

LLaMA model [57]. We find that the weights of each hidden unit have different normal distributions. As such, we test he weights of each individual hidden unit. This mean for weight $\mathbf{W} \in \mathcal{R}^{in \times out}$ we perform tests over the out dimension. Using a 5% significance threshold, we find that 7.5% of neurons are non-normally distributed which is about 2.5% more than the expected false-positive rate. As such, while almost all pretrained weights appear to be normally distributed there seem to be exceptions. Such exceptions might be due to outliers weights [13] or because the p-value of the Shaprio-Wilk test is not accurate for large samples sizes[53] that occur in the LLaMA FFN layer hidden units. this verifies the claim that neural network weights.

Table 12: Aggregated pairwise GPT-4 judgments between systems where the value of a cell at row x and column y is $\frac{\text{\# judgment } x \text{ is better than } y - \text{\# judgment } y \text{ is better than } x}{\text{total \# number of judgments}}$

Model	Guanaco 65B	Guanaco 33B	Vicuna	ChatGPT-3.5 Turbo	Bard	Guanaco 13B	Guanaco 7B
Guanaco 65B	-	0.21	0.19	0.16	0.72	0.59	0.86
Guanaco 33B	-0.21	-	0.17	0.10	0.51	0.41	0.68
Vicuna	-0.19	-0.17	-	0.10	0.50	0.20	0.57
ChatGPT-3.5 Turbo	-0.16	-0.10	-0.10	-	0.35	0.19	0.40
Bard	-0.72	-0.51	-0.50	-0.35	-	0.12	0.03
Guanaco 13B	-0.59	-0.41	-0.20	-0.19	-0.12	-	0.20
Guanaco 7B	-0.86	-0.68	-0.57	-0.40	-0.03	-0.20	-



Task

We would like to request your feedback on the performance of two AI assistants in response to the user question displayed below.

Please rate the helpfulness, relevance, accuracy, level of details of their responses. Each assistant receives an overall score on a scale of 1 to 10, where a higher score indicates better overall performance.

Please first rate each response out of 10.

Next, please provide a comprehensive explanation of your evaluation, avoiding any potential bias and ensuring that the order in which the responses were presented does not affect your judgment.

User Question

Imagine you are a time traveler from the year 3000. What technological advancements would you tell people about?

Response A

As an AI assistant, I don't have personal experiences or opinions. But I can tell you about some of the possible technological advancements that might exist in the year 3000 based on current trends and research. - Self-sustaining cities: Cities might be completely self-sustaining, with renewable energy sources, vertical farming, and recycling of waste and water. ...

Response B

As a time traveler from the year 3000, I would tell people about the following technological advancements: 1. Advanced Artificial Intelligence: In the future, AI is so advanced that it can completely automate many jobs that humans currently do. This has resulted in increased productivity and efficiency across many industries. ...

Rating for Response A

1
2
3
4
5
6
7
8
9
10

Rating for Response B

1
2
3
4
5
6
7
8
9
10

Comprehensive Explanation of Your Evaluation

Response X was better because...

Submit

图5：人工标注者使用的众包表单。

LLaMA 模型 [57]。我们发现每个隐藏单元的权重分布各不相同。因此，我们对每个隐藏单元的权重进行了测试。这意味着对于权重 $\mathbf{W} \in \mathcal{R}^{in \times out}$，我们在 out 维度上进行测试。使用 5% 的显著性阈值，我们发现 7.5% 的神经元不服从正态分布，这比预期的假阳性率高约 2.5%。因此，虽然几乎所有预训练权重看起来都服从正态分布，但似乎也存在例外。这些例外可能是由于异常权重 [13]，或者由于 Shaprio-Wilk 测试的 p 值对于 LLaMA FFN 层隐藏单元中出现的大样本量 [53] 不够准确。这验证了神经网络权重的相关说法。

表 12：系统间聚合的成对 GPT-4 判断，其中第 x 行、第 y 列的单元格值为 $\frac{\#\text{ judgment } x \text{ is better than } y - \#\text{ judgment } y \text{ is better than } x}{\text{total \# number of judgments}}$

Model	Guanaco 65B	Guanaco 33B	Vicuna	ChatGPT-3.5 Turbo	Bard	Guanaco 13B	Guanaco 7B
Guanaco 65B	-	0.21	0.19	0.16	0.72	0.59	0.86
Guanaco 33B	-0.21	-	0.17	0.10	0.51	0.41	0.68
Vicuna	-0.19	-0.17	-	0.10	0.50	0.20	0.57
ChatGPT-3.5 Turbo	-0.16	-0.10	-0.10	-	0.35	0.19	0.40
Bard	-0.72	-0.51	-0.50	-0.35	-	0.12	0.03
Guanaco 13B	-0.59	-0.41	-0.20	-0.19	-0.12	-	0.20
Guanaco 7B	-0.86	-0.68	-0.57	-0.40	-0.03	-0.20	-

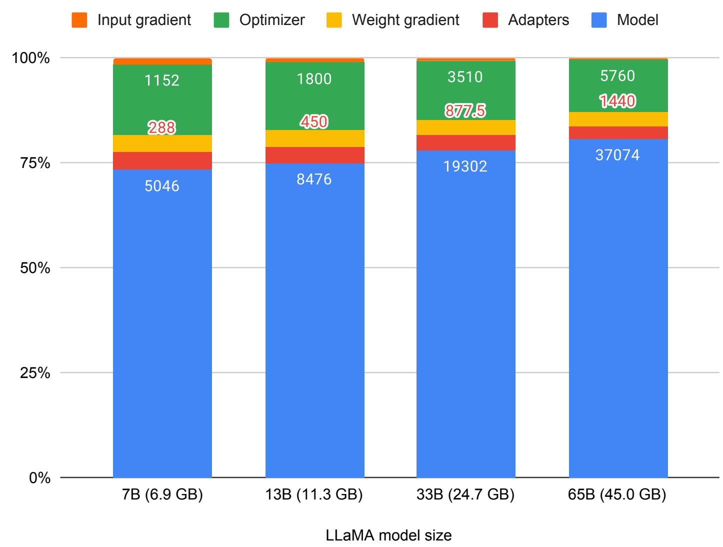


Figure 6: Breakdown of the memory footprint of different LLaMA models. The input gradient size is for batch size 1 and sequence length 512 and is estimated only for adapters and the base model weights (no attention). Numbers on the bars are memory footprint in MB of individual elements of the total footprint. While some models do not quite fit on certain GPUs, paged optimzier provide enough memory to make these models fit.

G Memory Footprint

The memory footpring for QLoRA training with different LLaMA base models can be seen in Figure 6. We see that the 33B model does not quite fit into a 24 GB and that paged optimizers are needed to train it. Depicted is also batch size 1 with a sequence length of 512 and gradient checkpointning. This means, if one uses a larger batch size, or if a long sequence is processed, the activation gradient might consume a considerable amount of memory.

Table 13: The complete ordering induced by pairwise GPT-4 judgments between systems

Model	Params	Size
Guanaco	65B	41 GB
Guanaco	33B	21 GB
Vicuna	13B	26 GB
ChatGPT-3.5 Turbo	N/A	N/A
Bard	N/A	N/A
Guanaco	13B	10 GB
Guanaco	7B	5 GB

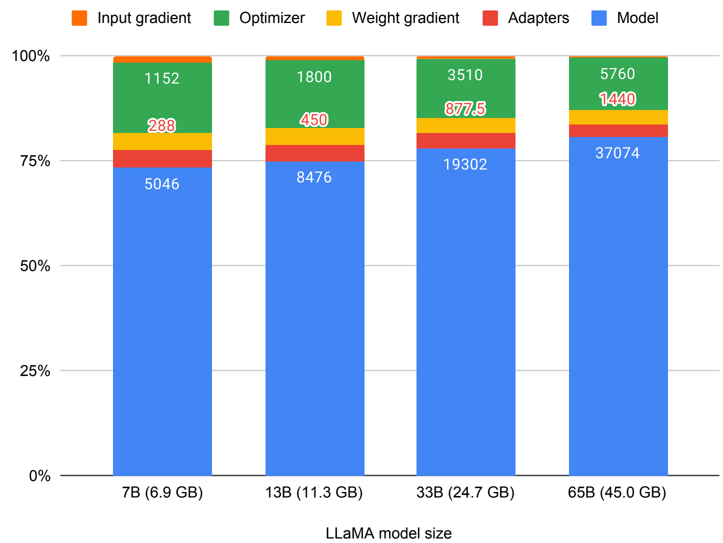


图6：不同LLaMA模型的内存占用细分。输入梯度大小是针对批量大小为1、序列长度为512的情况，仅估算适配器和基础模型权重（不包括注意力）。条形图上的数字表示总内存占用中各个元素的内存占用（以MB为单位）。虽然某些模型在某些GPU上可能无法完全载入，但分页优化器提供了足够的内存，使这些模型可以载入。

G 内存占用

使用不同 LLaMA 基础模型进行 QLoRA 训练的内存占用情况见图 6。我们可以看到，33B 模型在 24 GB 内存下无法完全适配，因此需要使用分页优化器来训练它。图中还展示了批量大小为 1，序列长度为 512，并使用了梯度检查点。这意味着，如果使用更大的批量大小，或者处理较长的序列，激活梯度可能会消耗大量内存。

表 13：由系统之间成对 GPT-4 判断引起的完整排序

Model	Params	Size
Guanaco	65B	41 GB
Guanaco	33B	21 GB
Vicuna	13B	26 GB
ChatGPT-3.5 Turbo	N/A	N/A
Bard	N/A	N/A
Guanaco	13B	10 GB
Guanaco	7B	5 GB